Courier Management System (CMS)

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DEPARTMENT : BSC.COMPUTER SCIENCE

COLLEGE CODE : TVU304

OUR TEAM MEMBERS :

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User Story:

Courier operations become difficult to control when booking details, parcel movement, delivery assignments and billing records are maintained in separate places. Staff may spend time checking registers or messages, customers may not receive timely shipment updates, and managers may lack a consolidated view of branch performance. The proposed Courier Management System provides one Salesforce-based workspace for registering customers, creating shipments, assigning delivery agents, recording delivery events, preparing invoices and monitoring operational reports. The aim is to make the process traceable from booking to final delivery while reducing repetitive manual work.

1. INTRODUCTION

1.1 Project Overview

Courier Management System (CMS) is a Salesforce-based application designed to organize the complete lifecycle of a courier shipment. It connects customer information, shipment records, branch operations, delivery-agent activity, delivery updates and invoice information in a

common data model. Users can create and locate shipment records, maintain delivery status, review outstanding work and obtain management information through Salesforce reports and dashboards.

1.2 Purpose

The purpose of CMS is to provide a consistent digital workflow for courier booking and delivery operations. Instead of maintaining separate registers for customers, parcels, agents and payments, the application stores related information in Salesforce objects and uses validation, automation and role-based access to improve data quality. The system is intended to support timely updates, clearer accountability and easier operational reporting.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	coordinate daily courier activity	assignments and updates are spread across different records	there is no single operational view	overloaded and uncertain
PS-2	Customer	know where my parcel is	status information may not be updated promptly	tracking depends on manual communication	anxious about delivery
PS-3	Delivery Agent	receive clear delivery work	job details can be communicated through multiple channels	there is no structured assignment queue	confused and delayed



2.2 Empathy Map Canvas

User Persona: Branch Manager

- Think & Feel: Wants every shipment to have a clear owner, status and next action.
- Hear: Requests for delivery updates, customer follow-ups and agent availability information.
- See: Shipment records, branch workload, pending deliveries and exception cases.
- Say & Do: Checks assignments, follows delayed shipments and coordinates staff.
- Pain: Missing updates, duplicate entries, delayed hand-offs and billing discrepancies.
- Gain: A unified workspace, automatic status synchronization and summary dashboards.



2.3 Brainstorming & Idea Prioritization

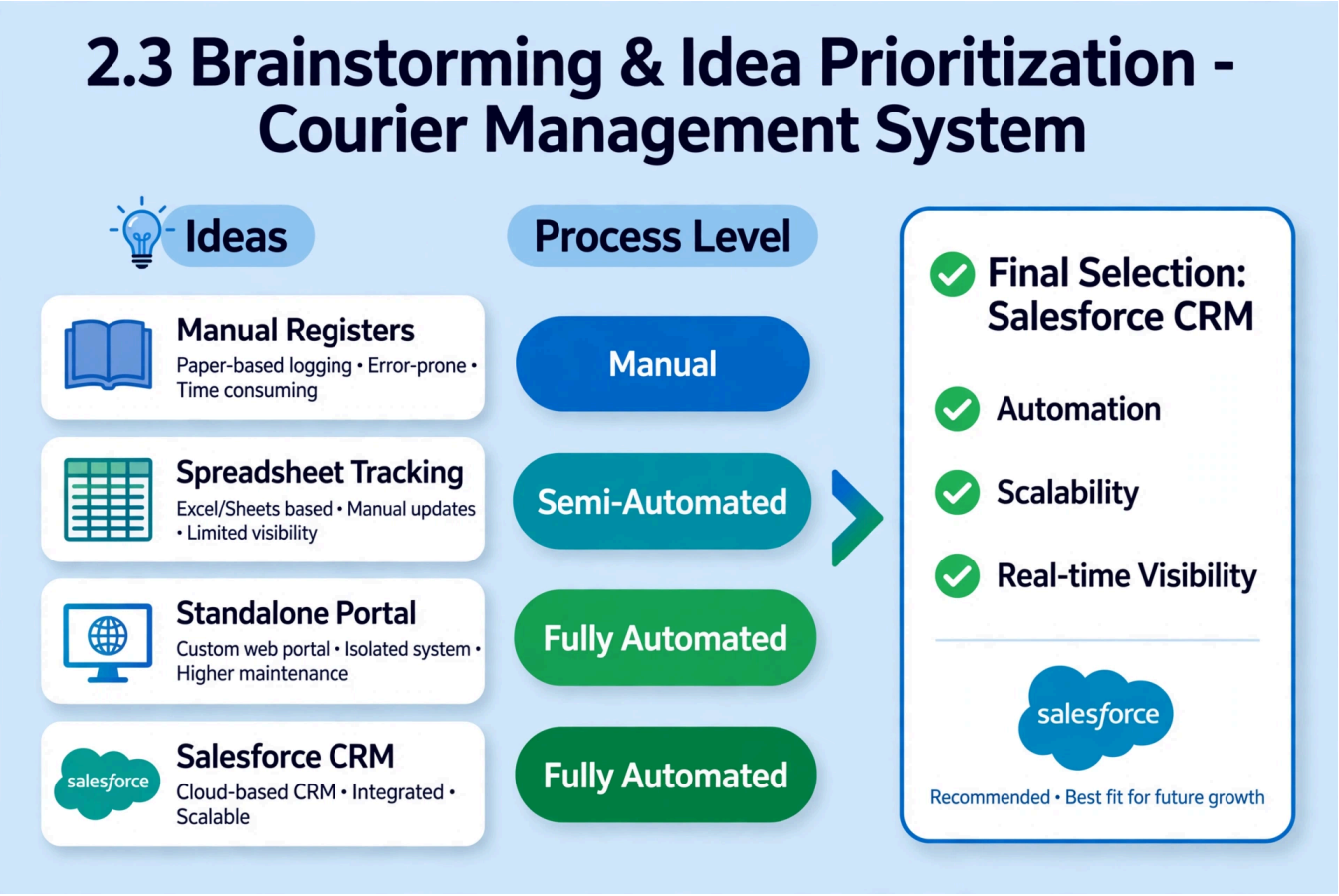
Ideas Considered

- Paper/register based tracking
- Shared spreadsheet workflow
- Standalone web portal
- Salesforce-based courier application

Final Selection

A Salesforce implementation was selected for the project because the template workflow requires centralized records, configurable automation, access control, reporting and an application interface that can be extended as operational needs grow.

2.3 Brainstorming & Idea Prioritization - Courier Management System



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

● **Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting**



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Create and maintain customer records.
FR-2	Register shipments with service and destination details.
FR-3	Assign a delivery agent to an active shipment.
FR-4	Record and synchronize shipment status updates.
FR-5	Create invoice records linked to shipments.
FR-6	Provide operational reports and dashboards.

Non-Functional Requirements

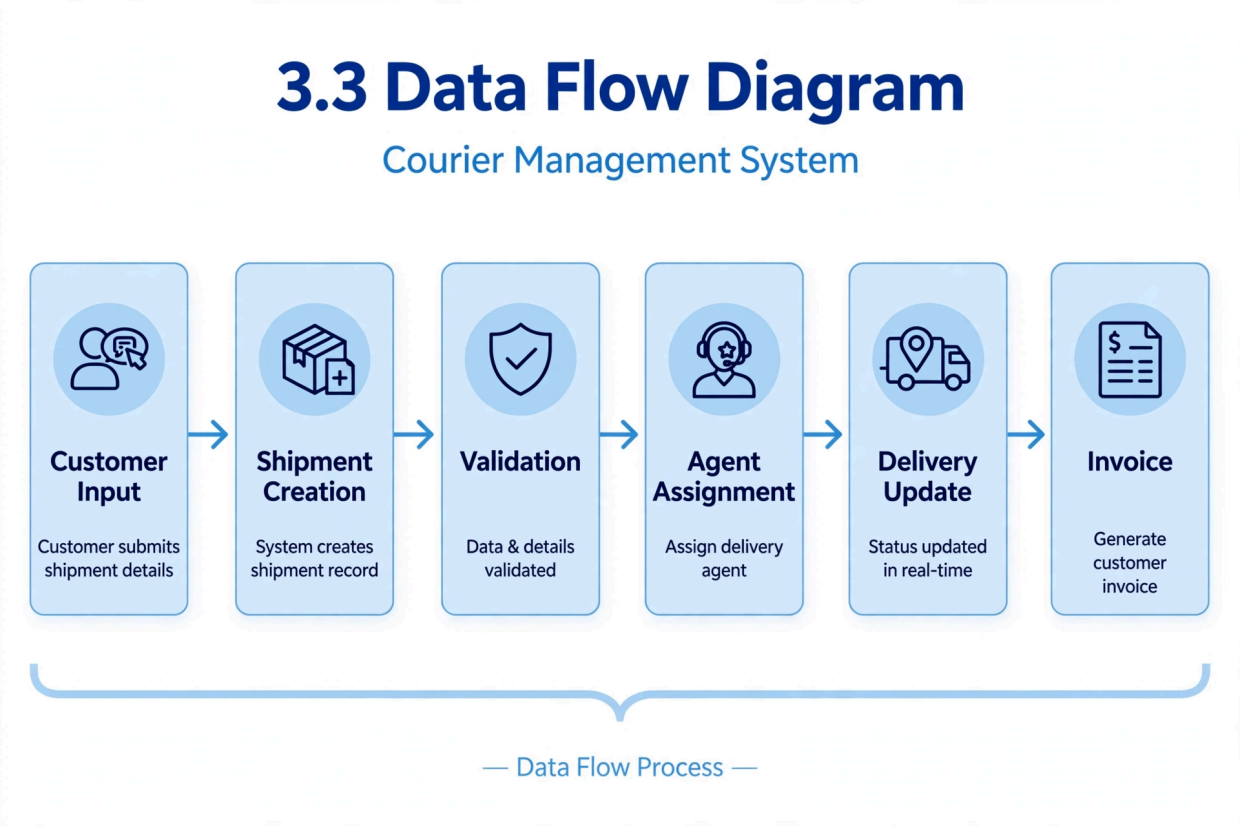
NFR	Description
Usability	Provide a straightforward Lightning interface for routine staff tasks.

Security	Restrict records and actions according to user responsibilities.
Reliability	Prevent invalid operational records through validation and controlled workflows.
Performance	Allow users to locate and update normal shipment records without unnecessary steps.
Availability	Use Salesforce cloud access for authorized users.
Scalability	Allow additional customers, branches, agents and shipments to be added without redesigning the whole process.

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports



3.4 Technology Stack

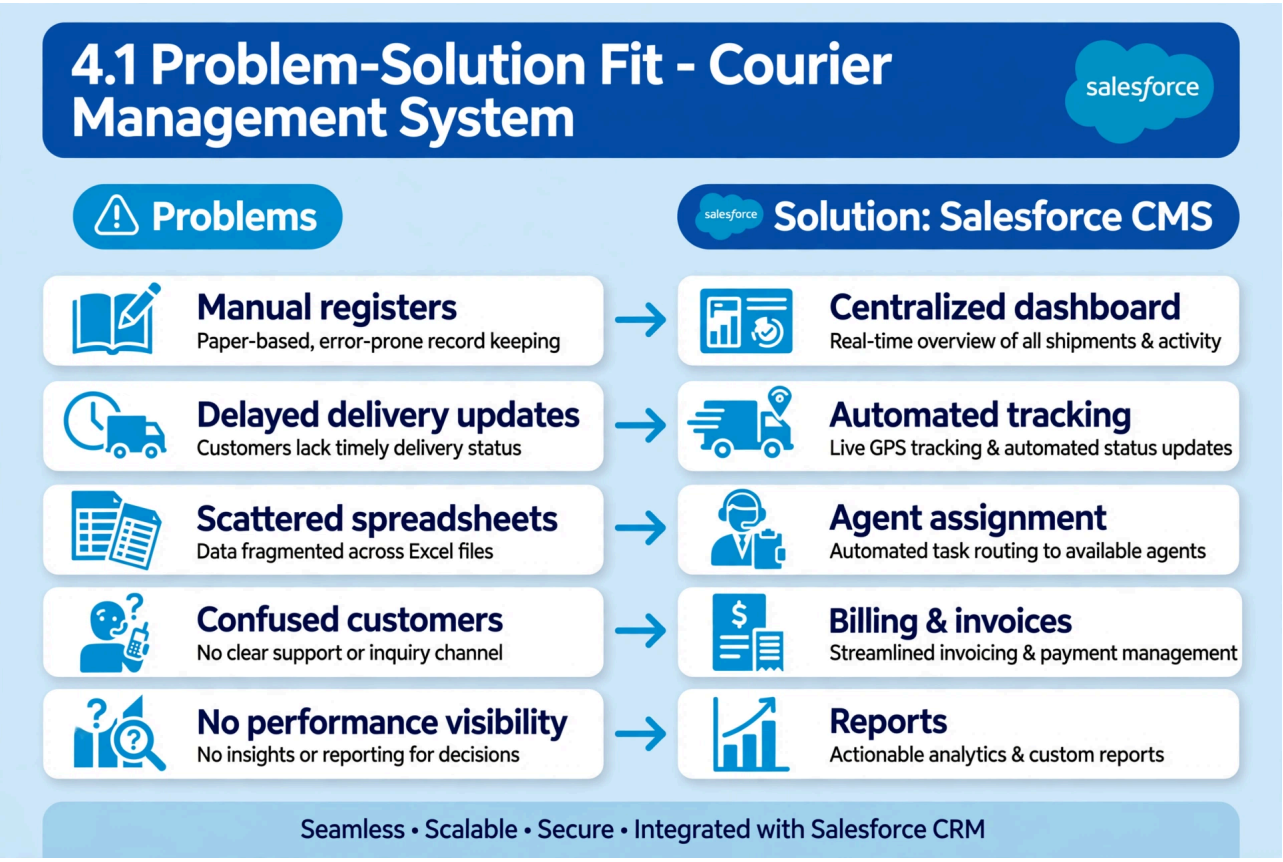
Layer	Technology
UI	Salesforce Lightning Experience
Logic	Salesforce Flow
Database	Salesforce Custom Objects
Automation	Record-triggered Flows and Validation Rules

Security	Profiles, Roles and Object Permissions
Reporting	Salesforce Reports and Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The design connects each major operational problem with a Salesforce capability. Fragmented customer and parcel information is addressed through related objects; inconsistent shipment data is controlled by validation rules; repetitive status handling is supported by Flow automation; and management visibility is improved through reports and dashboards. The design therefore follows the actual courier lifecycle instead of treating booking, delivery and billing as unrelated activities.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Courier information is difficult to follow when activities are recorded independently.
Solution	A Salesforce application that connects customer, shipment, agent, branch, update and invoice records.
Innovation	Automated operational updates and a single view of related shipment information.

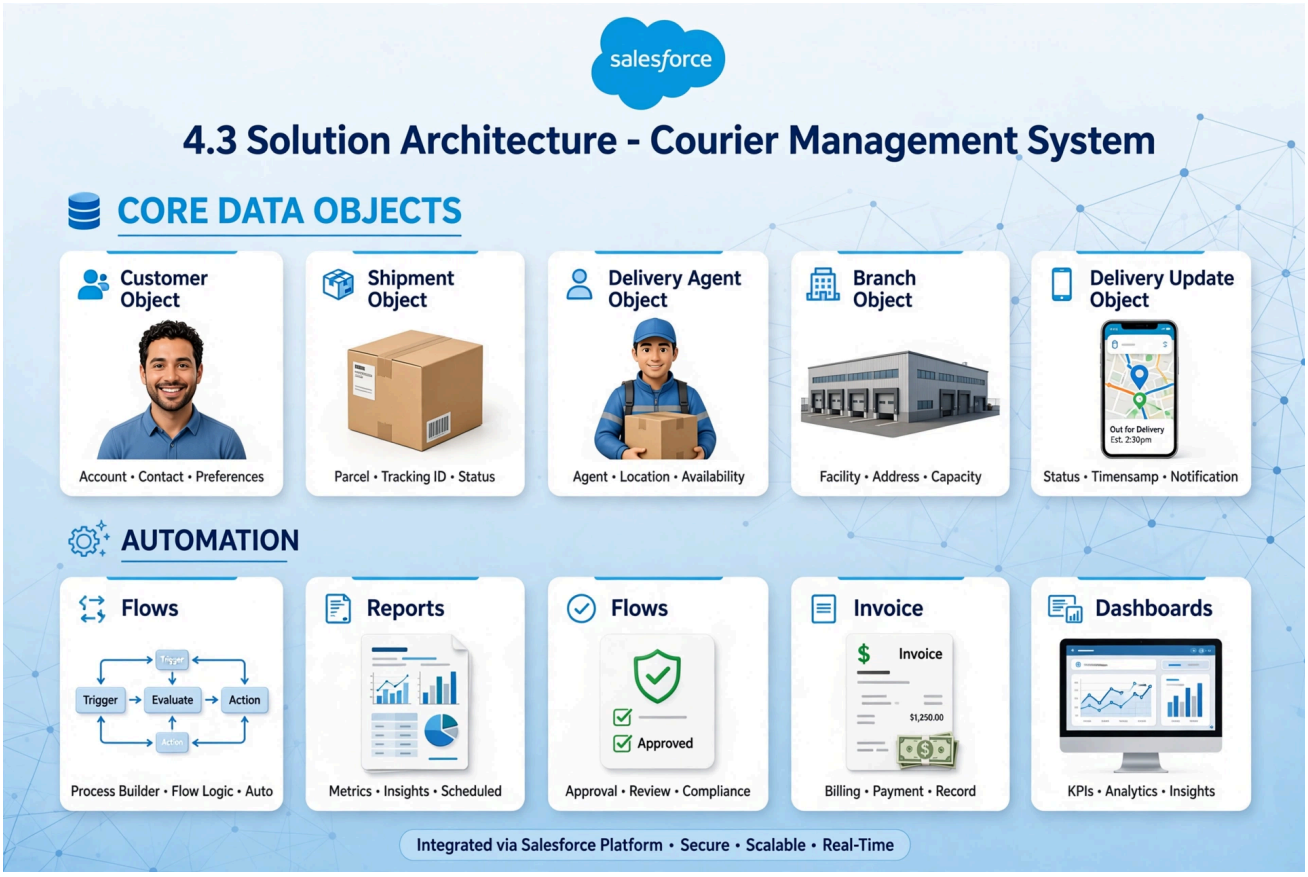
Impact	Less repetitive entry and faster identification of pending work.
Business Model	Supports routine courier booking, delivery coordination and billing operations.
Scalability	New branches, agents, services and records can be introduced through the Salesforce data model.

4.3 Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Validation Rules
- Record-triggered Flows
- Reports
- Dashboards

Relationship flow: Customer → Shipment → Delivery Updates; Shipment → Delivery Agent; Shipment → Branch; Shipment → Invoice.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → User Stories → Tasks → Story Points
- Sprint-based execution
- Velocity used to review completed work and plan subsequent sprints

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Platform Setup	USN-1	As an administrator, I want a configured	3	High	Member 1

			Salesforce environment so that CMS development can begin.			
Sprint-2	Fields & Relationships	USN-3	As an admin, I want meaningful fields and relationships so that shipment information is connected correctly.	5	High	Member 1
Sprint-3	Validation	USN-4	As an operator, I want invalid shipment and delivery data to be blocked so that records remain dependable.	3	High	Member 2
Sprint-4	Security & App	USN-6	As an administrator, I want appropriate profiles, roles and application navigation so that users see relevant functions.	5	Medium	Member 2
Sprint-5	Reporting	USN-7	As a manager, I want reports and dashboards so that courier activity can be reviewed.	3	Medium	Member 1

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	**Planned Points**	**Completed Points**	**Velocity**	**Status**
Sprint-1	3	3	3	Completed
Sprint-2	5	4	4	Completed
Sprint-3	3	3	3	Completed
Sprint-4	5	4	4	Completed
Sprint-5	3	3	3	Completed

Burndown interpretation: the remaining work is tracked by comparing planned story points with completed points at the end of each sprint. Any unfinished item is carried into the next planning cycle rather than being treated as completed.

6. Project Development Phase:

What you'll learn

- Creating and configuring a Salesforce Developer environment
- Designing custom objects for courier operations
- Creating tabs, fields and relationships
- Applying validation rules
- Designing page layouts
- Building record-triggered flows
- Creating a Lightning application
- Configuring profiles and roles
- Creating users
- Building reports and dashboards

Milestone 1 - Salesforce Account:

Introduction

The first development activity is to establish the Salesforce environment used for CMS configuration and testing.

What Is Salesforce?

Salesforce is a cloud platform that provides configurable data, user interface, automation, security and reporting capabilities. In this project it is used as the application platform for the Courier Management System.

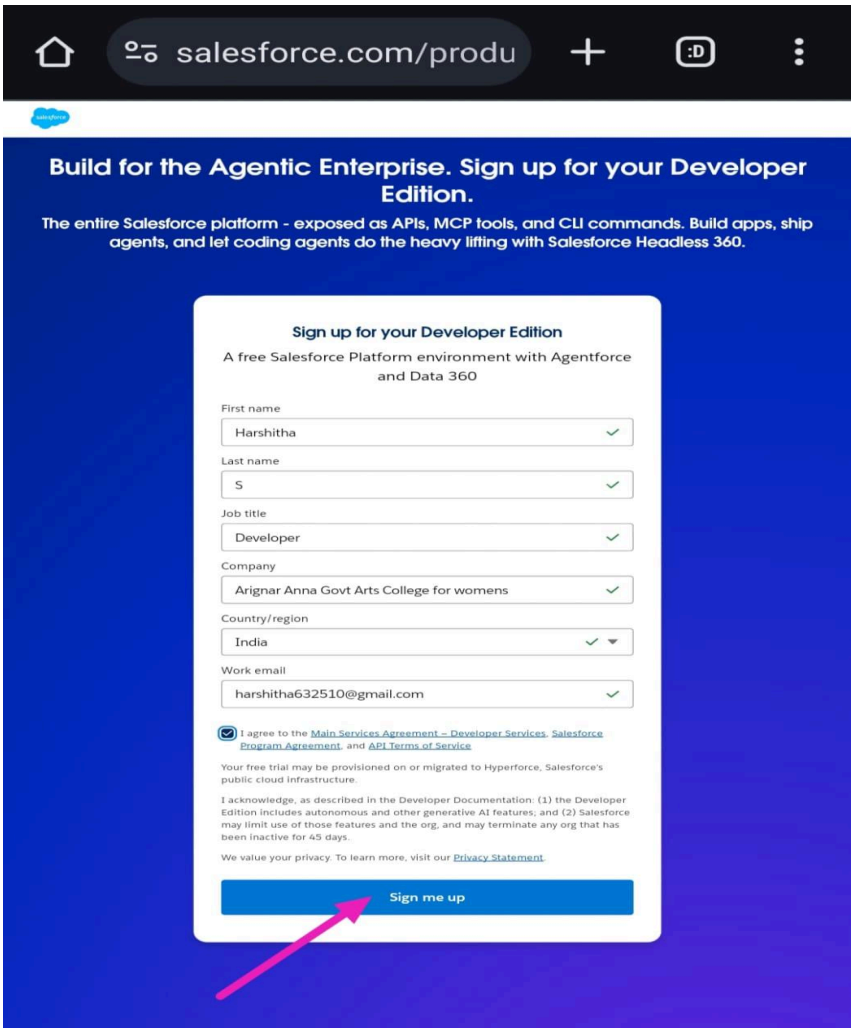
Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :

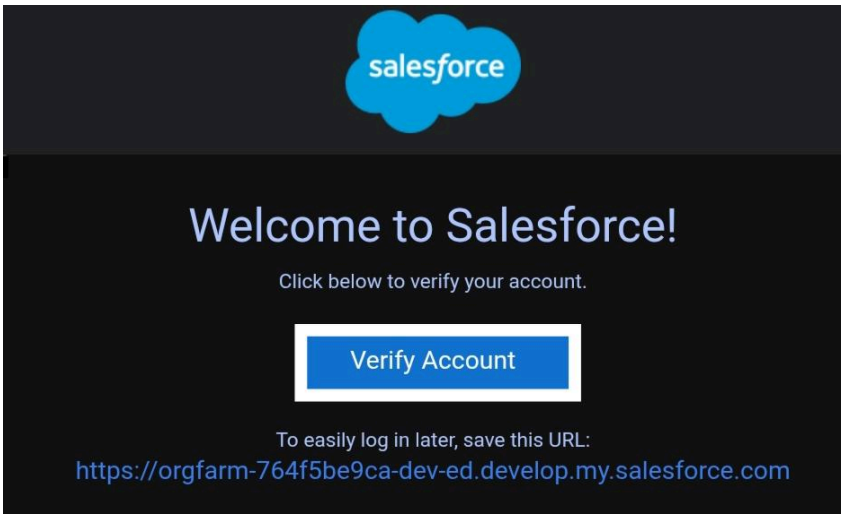
- First name & Last name
- Email
- Job Title: Developer
- Company: College Name
- Country: India

Click on sign me up after filling these.



Activity 2: Account Activation

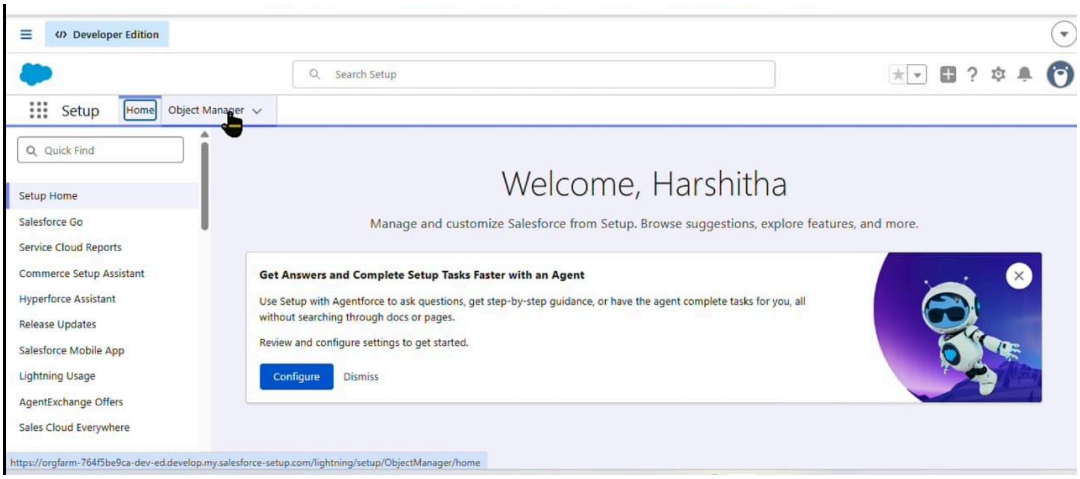
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account.

3.Give a password and answer a security question and click on change password.

4.Then you will redirect to your salesforce setup page.



Milestone 2- Object:

What is an object?

An object is a Salesforce data structure used to store a category of records. CMS uses custom objects to separate operational information while relationships connect records that belong to the same shipment process.

Custom Objects Created

Object	Purpose	Record Name
Customer__c	Stores customer contact and service information.	Customer Name

Shipment__c	Stores parcel booking and shipment lifecycle information.	Shipment Number
Delivery_Agent__c	Stores delivery staff information.	Agent Name
Branch__c	Stores branch information.	Branch Name
Delivery_Update__c	Stores individual tracking events.	Update Number
Invoice__c	Stores shipment billing information.	Invoice Number

Relationships

Shipment__c is the central operational object. It is related to Customer__c, Delivery_Agent__c and Branch__c. Delivery_Update__c is linked to Shipment__c so that each status event belongs to a shipment. Invoice__c is linked to Shipment__c so billing can be traced back to the corresponding parcel.

Steps to Implement

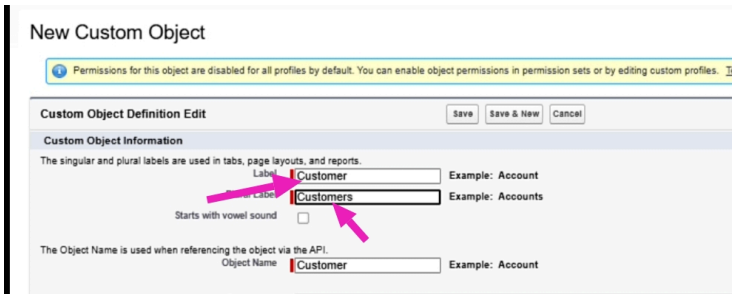
Activity 1: Create Customer Object

1. Click Setup → Object Manager → Create → Custom Object



2. Enter the following details:

- Label: Customer
- Plural Label: Customers
- Record Name: Customer Name (Text)



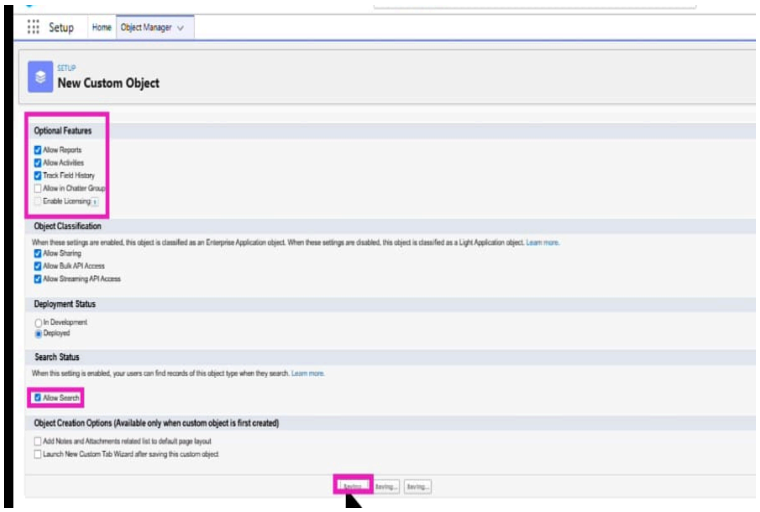
3. Under Optional Features:

- Enable Allow Reports
- Enable Track Field History

● In Search Status check the box Allow Search

● In Object Creation Options check the box Add Notes and Attachments related list to default page layout

4. Click Save.



Activity 2: Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:

● Label: Shipment

● Plural Label: Shipments

● Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}

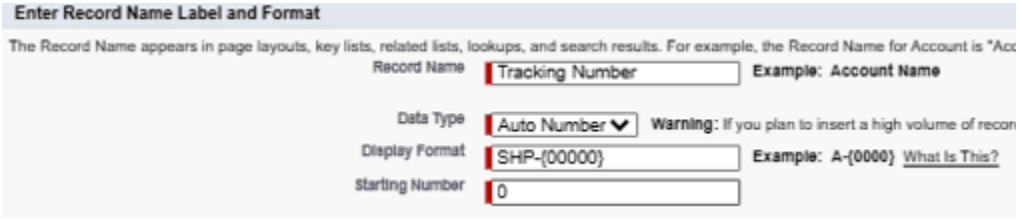
Starting Number: 0

4. Enable:

● Allow Reports

● Track Field History

5. Click Save



Activity 3: Create Delivery Agent Object

1. Setup → Object Manager → Create → Custom Object

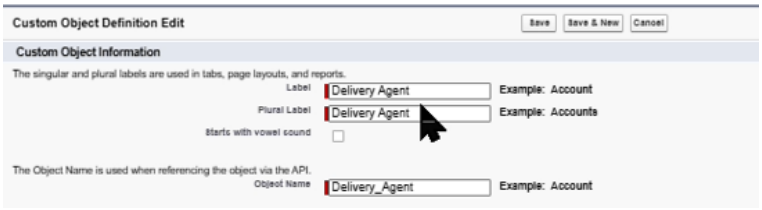
2. Enter:

- Label: Delivery Agent
- Plural Label: Delivery Agents
- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports
- Track Field History

4. Click Save





Activity 4: Create Branch Object

1. Setup → Object Manager → Create → Custom Object

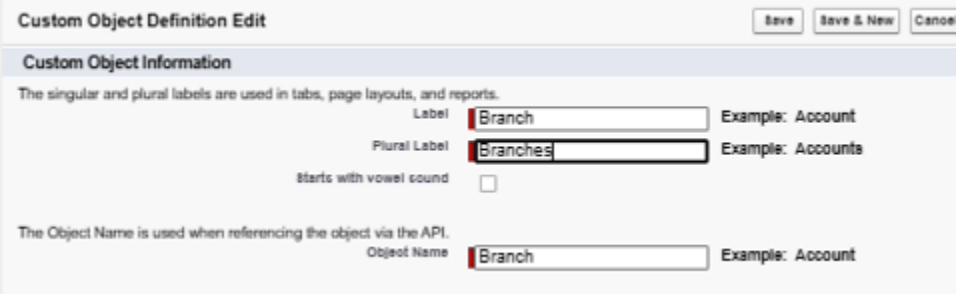
2. Enter:

- Label: Branch
- Plural Label: Branches
- Record Name: Branch Name (Text)

3. Enable:

- Allow Reports

4. Click Save.



Activity 5: Create Delivery Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)

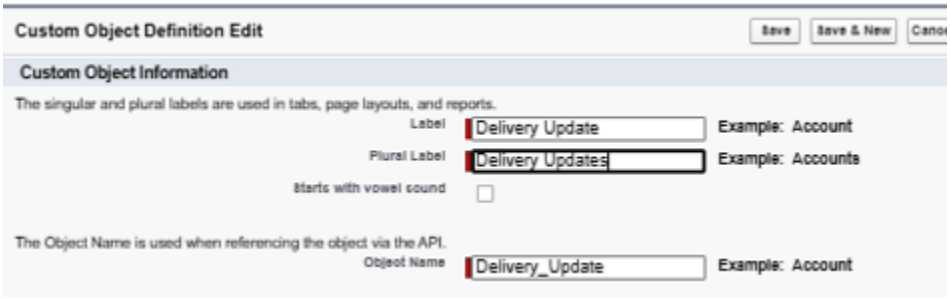
3. Example Format: UPD-{0000}

Enter Starting number : 0

4. Enable:

- Allow Reports
- Track Field History

5. Click Save.



Activity 6: Create Invoice Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

● Label: Invoice

● Plural Label: Invoices

● Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}

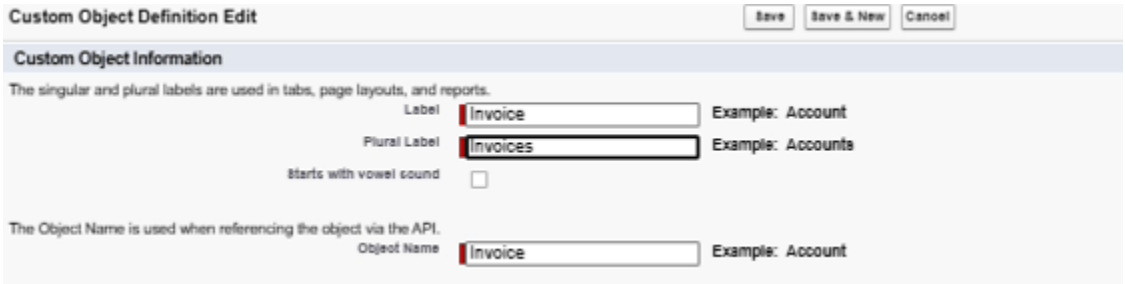
   Enter Starting number: 0

4. Enable:

  ● Allow Reports

  ● Track Field History

5. Click Save



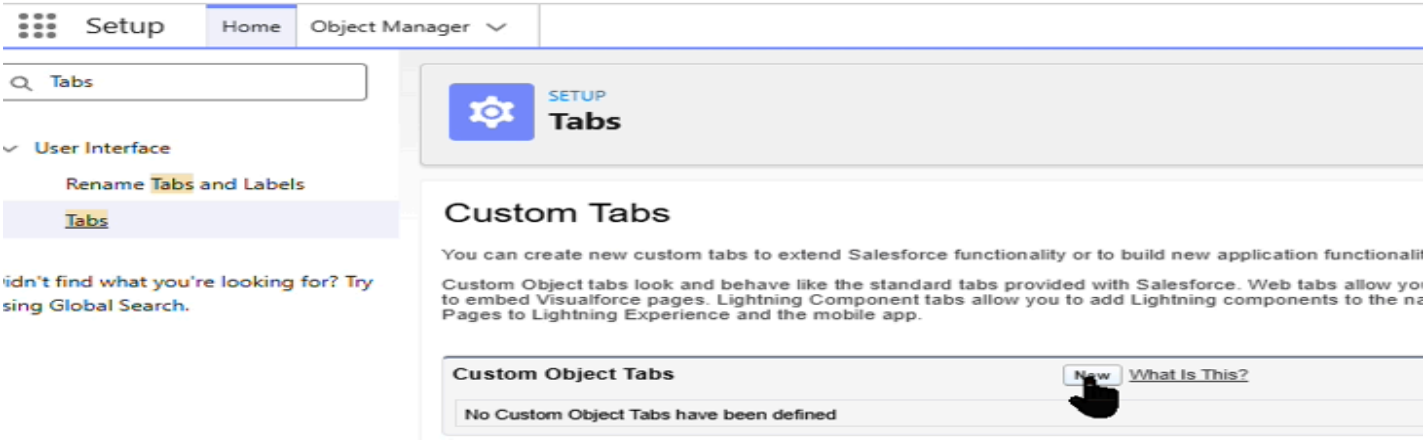
Milestone 3 - Tabs:

Tab: Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

2. Under Custom Object Tabs, click New.

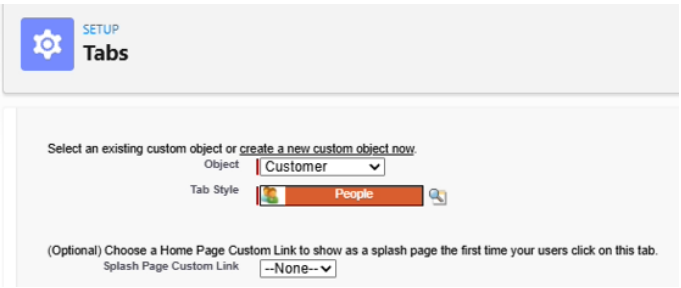


3. For Object, select Customer.

4. For Tab Style, choose any icon.

5. Leave all default settings as they are.

6. Click Next, Next, and Save.



Activity 2: Creating a Tab for Shipment Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

2. Under Custom Object Tabs, click New.

3. For Object, select Shipment.

4. Choose any Tab Style.

5. Keep defaults unchanged.

6. Click Next, Next, and Save.



Activity 3: Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

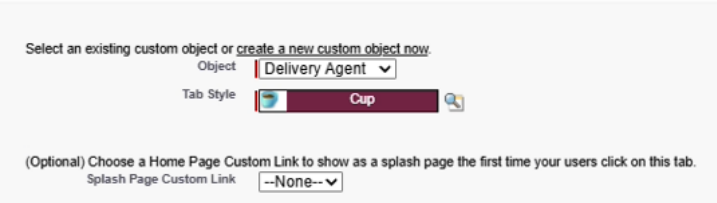
2. Under Custom Object Tabs, click New.

3. For Object, select Delivery Agent.

4. Pick any Tab Style.

5. Leave default settings.

6. Click Next, Next, and Save.



Activity 4: Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

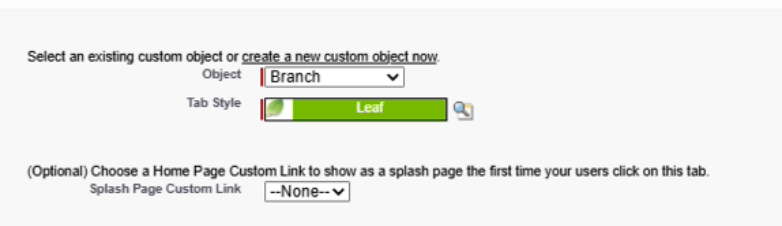
2. Under Custom Object Tabs, click New.

3. For Object, select Branch.

4. Choose any Tab Style.

5. Keep all defaults.

6. Click Next, Next, and Save.



Activity 5: Creating a Tab for Delivery Updates Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

2. Under Custom Object Tabs, click New.

3. For Object, select Delivery Update.

4. Select any Tab Style.

5. Keep default configurations.

6. Click Next, Next, and Save.

Activity 6: Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

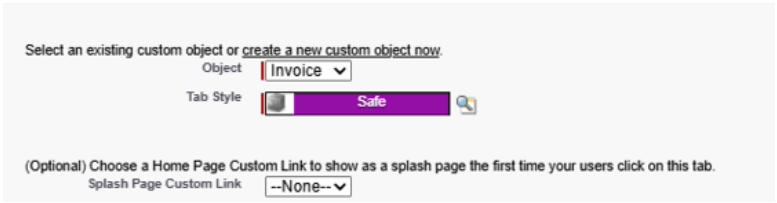
2. Under Custom Object Tabs, click New.

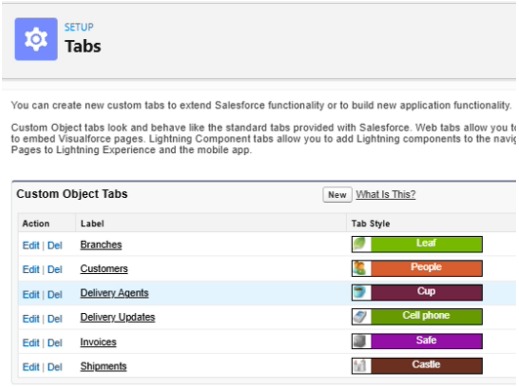
3. For Object, select Invoice.

4. Select any Tab Style.

5. Leave defaults unchanged.

6. Click Next, Next, and Save.





Milestone 4 - Fields & Relationships:

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1: Creation of fields for the Customer Object

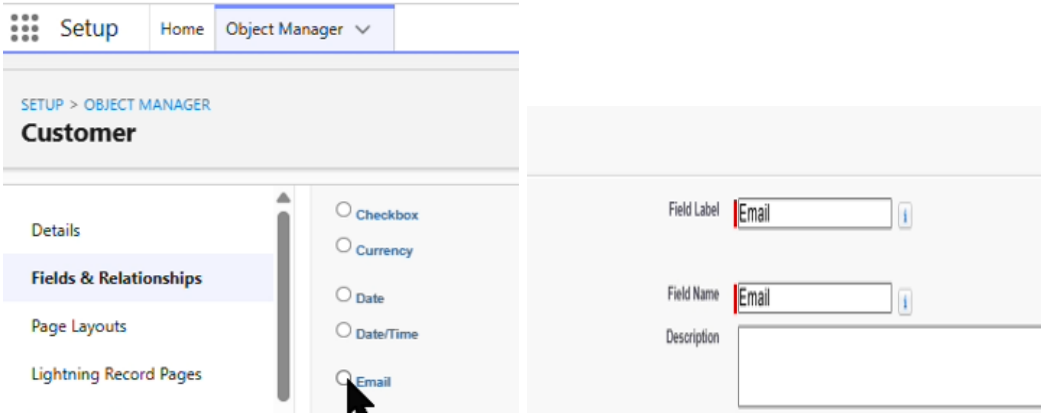
Field Name	API Name	Data Type	Required	Description
Customer Name	Customer_Name_c	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address

Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state

Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

1. Email (Email Field)

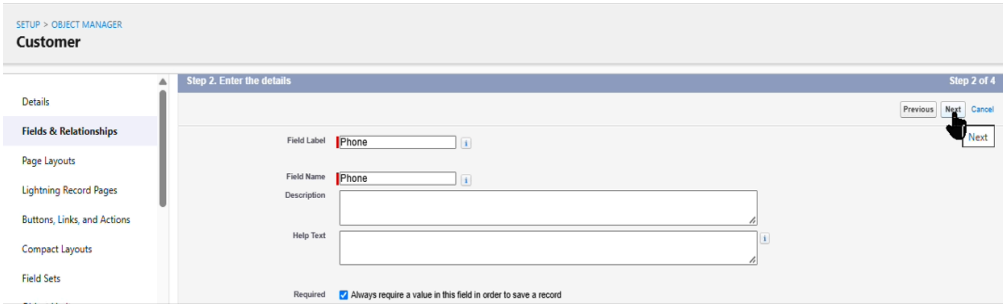
- Go to Setup → Object Manager → Customer__c
- Click Fields & Relationships → New
- Select Email → Next
- Field Label: Email
- Check the Required box
- Click Next → Next → Save



2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Phone → Next
- Field Label: Phone
- Check Required

● Click Next → Next → Save



3. Address (Text Area)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required
- Save



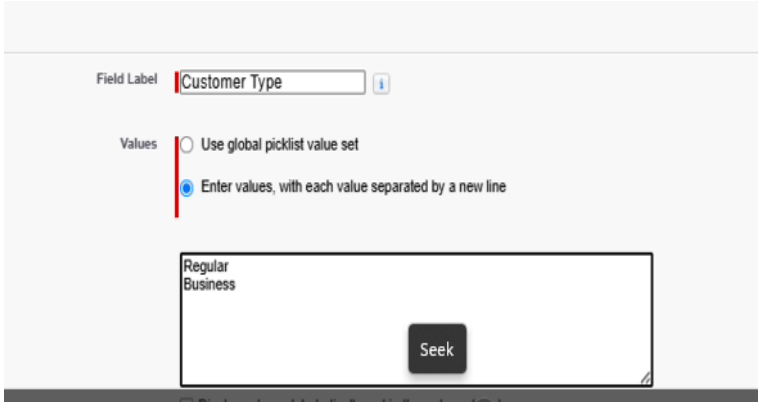
4. Customer Type (Picklist)

●Setup →Customer__c

→ Fields → New

● Picklist → Next

- Field Label: Customer Type
- Enter values:
    - Regular
    - Business
- Check Required
- Save



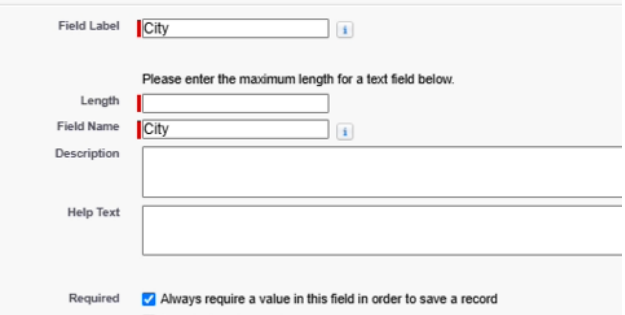
5. City (Text)

- Setup→Customer__c

→ Fields → New

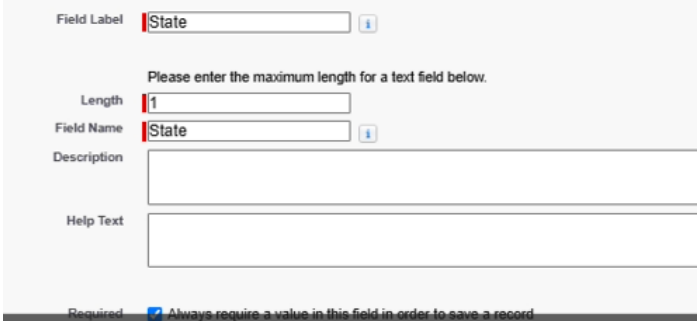
- Text → Next

- Field Label: **City**
- Save



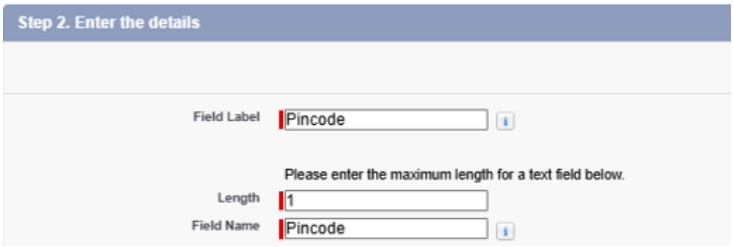
6. State (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **State**
- Save



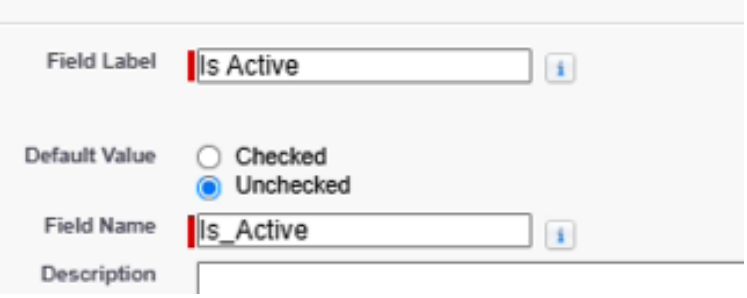
7. Pincode (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **Pincode**
- Save



8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
- Checkbox → Next
- Field Label: **Is Active**
- Save



Activity 2: Creation of fields for the Shipment Object

Field Name	API Name	Data Type	Required	Description

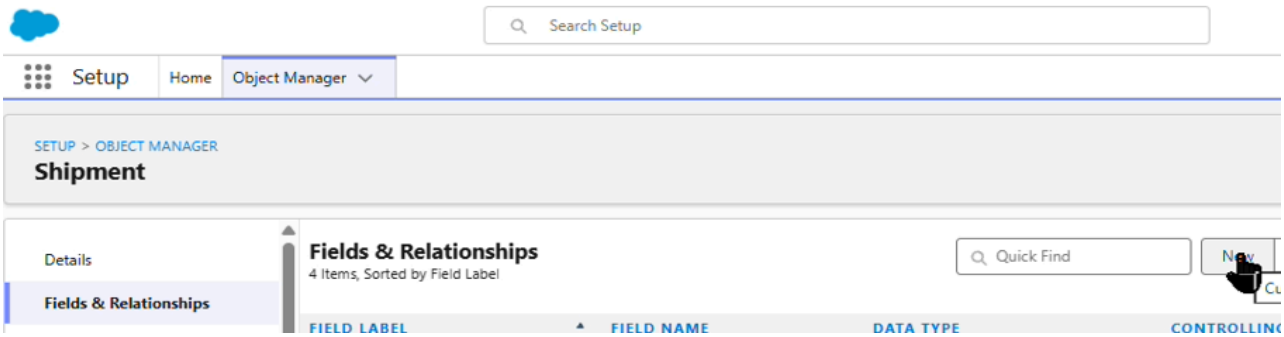
Tracking Number Name Auto Number Yes Unique ID

Customer	Customer__c	Lookup(Customer)	Yes	Related customer
Source Address	Source_Address__c	Text Area	Yes	Pickup location
Destination Address	Destination_Address_ _c	Text Area	Yes	Delivery location
Weight	Weight__c	Number	Yes	Parcel weight

Price	Price__c	Currency	Yes	Shipping charge
Status	Status__c	Picklist	Yes	Shipment status
Expected Date	Expected_Date__c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Agent__c	Lookup(Delivery_Age nt)	No	Delivery agent
Shipment Type	Shipment_Type__c	Picklist	Yes	Domestic/International

1. Source Address (Text Area)

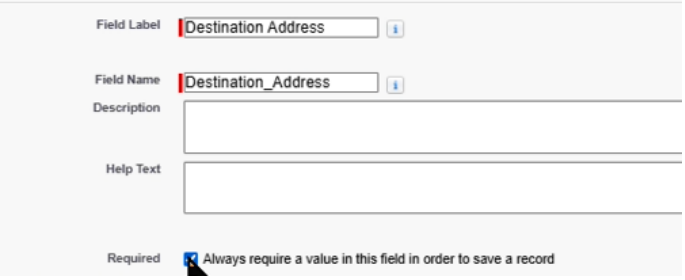
- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Source Address**
- Required → Save





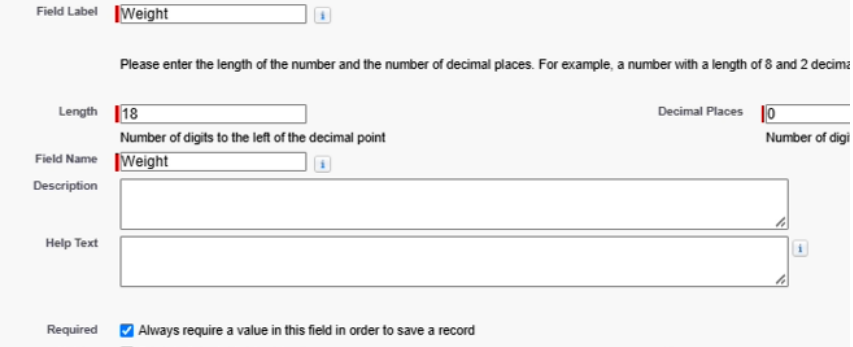
2. Destination Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Destination Address**
- Required → Save



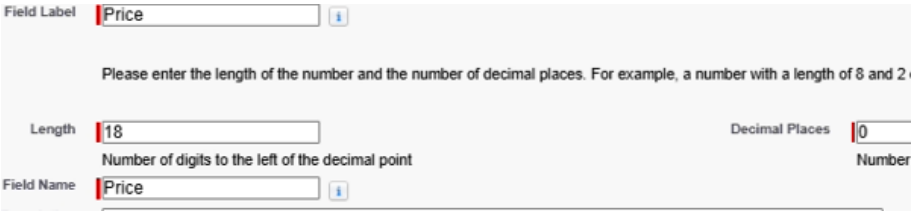
3. Weight (Number)

- Setup → Shipment__c → Fields → New
- Number → Next
- Field Label: **Weight**
- Required → Save



4. Price (Currency)

- Setup → Shipment__c → Fields → New
- Currency → Next
- Field Label: **Price**
- Required → Save

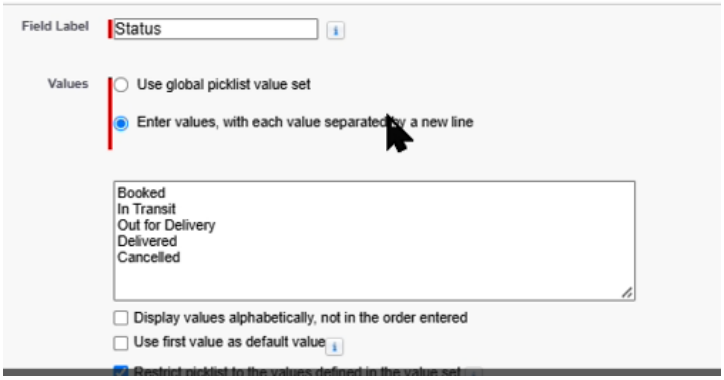


5. Status (Picklist)

Values:

- Booked
- In Transit
- Out for Delivery
- Delivered
- Cancelled

Required → Save



6. Expected Date (Date)

- Setup → Shipment__c → Fields → New
- Date → Next
- Field Label: **Expected Date**
- Required → Save

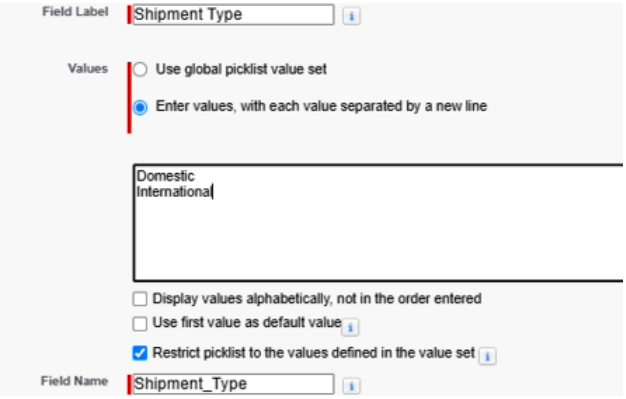


7. Shipment Type (Picklist)

Values:

- Domestic
- International

Required → Save



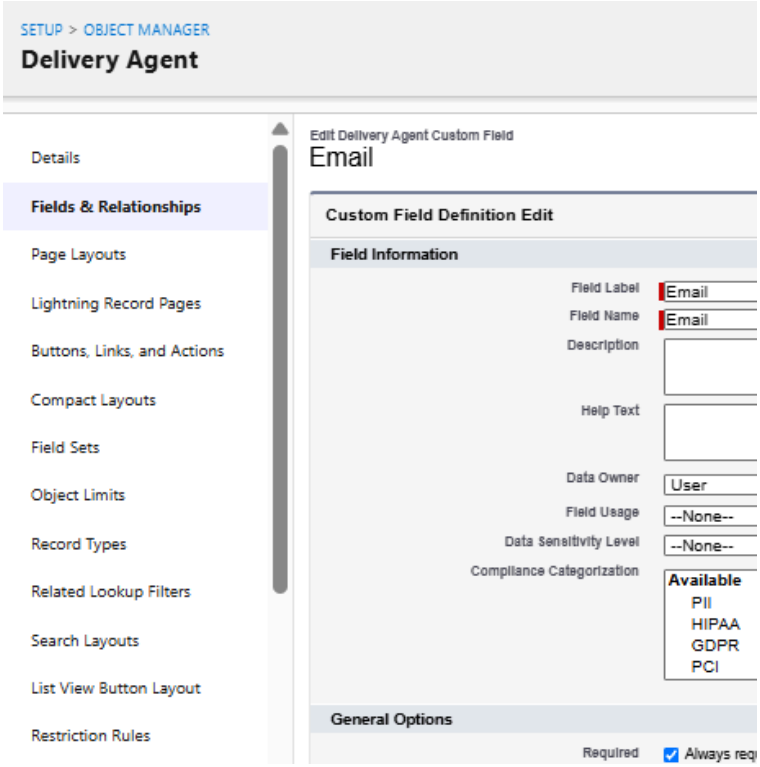
Activity 3: Creation of fields for Delivery Agent Object

Field Name	API Name	Data Type	Required
Agent Name Name Email Phone	Email__c Phone__c	Text Email Phone	Yes Yes Yes
Branch	Branch__c	Lookup	Yes

Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

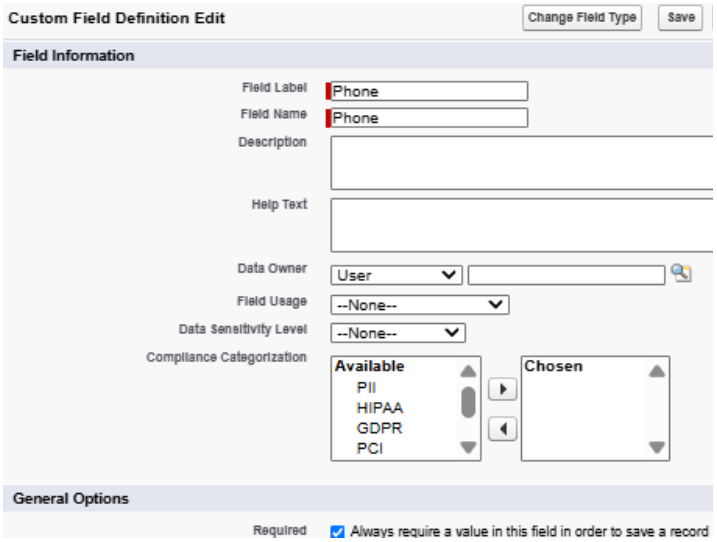
1. Email

- Setup → Delivery_Agent__c → Fields → New
- Email → Next
- Field Label: Email
- Required → Save



2. Phone

- Setup → Delivery_Agent__c → Fields → New
- Phone → Next
- Field Label: Phone
- Required → Save

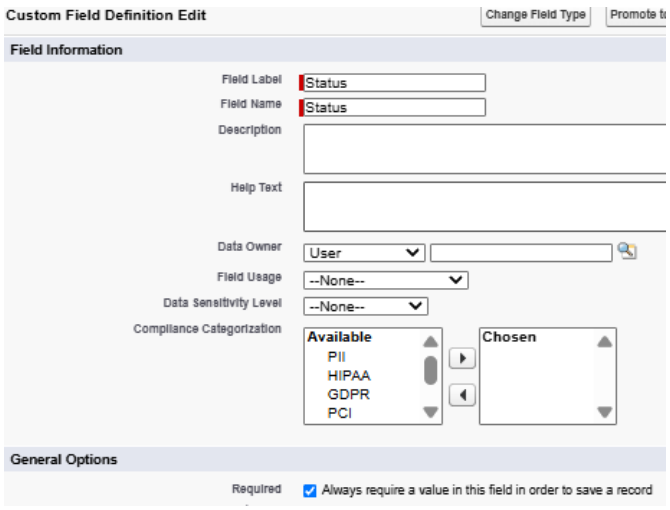


3. Status (Picklist)

Values:

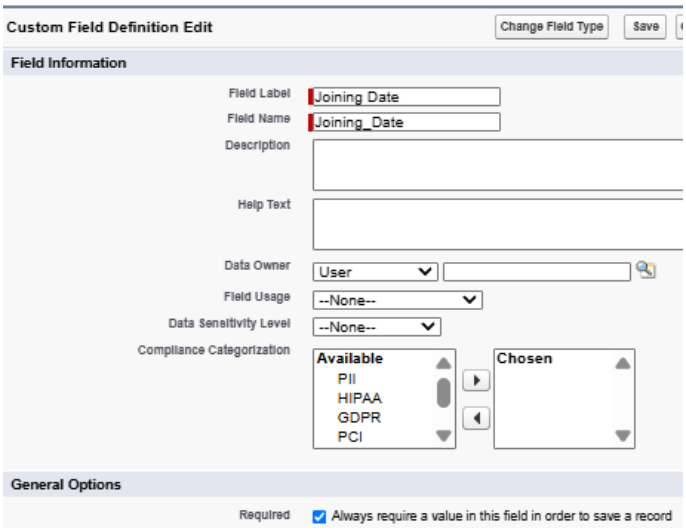
- Active
- Inactive
- On Leave

Required → Save



4. Joining Date

- Setup → Delivery_Agent__c → Fields → New
- Date → Next
- Field Label: Joining Date
- Save

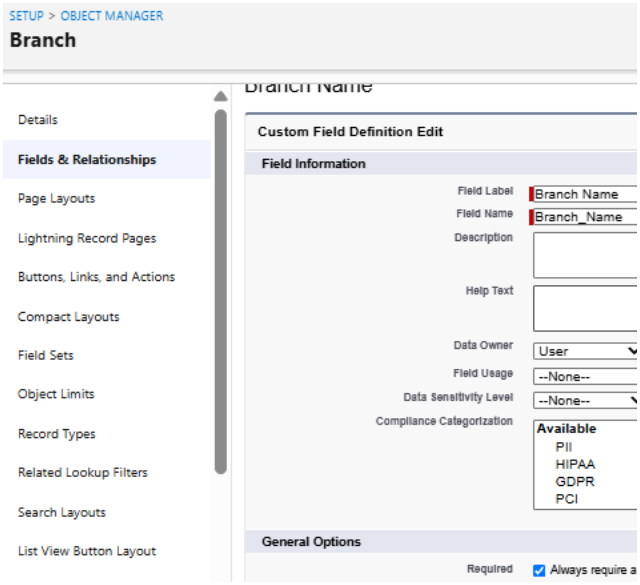


Activity 4: Creation of fields for the Branch Object

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes

Create each field using:

- Setup → Branch__c → Fields → New
- Select respective Data Type
- Enter Field Label
- Save



(Repeat for all fields)

Branch Code	Branch_Code__c	Text(1)
Branch Name	Branch_Name__c	Text(1)
Branch Name	Name	Text(80)
City	City__c	Text(1)
Created By	CreatedById	Lookup(User)
Last Modified By	LastModifiedById	Lookup(User)
Manager Name	Manager_Name__c	Text(1)
Owner	OwnerId	Lookup(User,Group)
State	State__c	Text(1)

Branch Code	Branch_Code__c	Text(1)
Branch Name	Branch_Name__c	Text(1)
Branch Name	Name	Text(80)
City	City__c	Text(1)
Created By	CreatedById	Lookup(User)
Last Modified By	LastModifiedById	Lookup(User)
Manager Name	Manager_Name__c	Text(1)
Owner	OwnerId	Lookup(User,Group)
State	State__c	Text(1)

Branch Code	Branch_Code__c	Text(1)
Branch Name	Branch_Name__c	Text(1)
Branch Name	Name	Text(80)
City	City__c	Text(1)
Created By	CreatedById	Lookup(User)
Last Modified By	LastModifiedById	Lookup(User)
Manager Name	Manager_Name__c	Text(1)
Owner	OwnerId	Lookup(User,Group)
State	State__c	Text(1)

Activity 5: Creation of fields for the Delivery Update Object

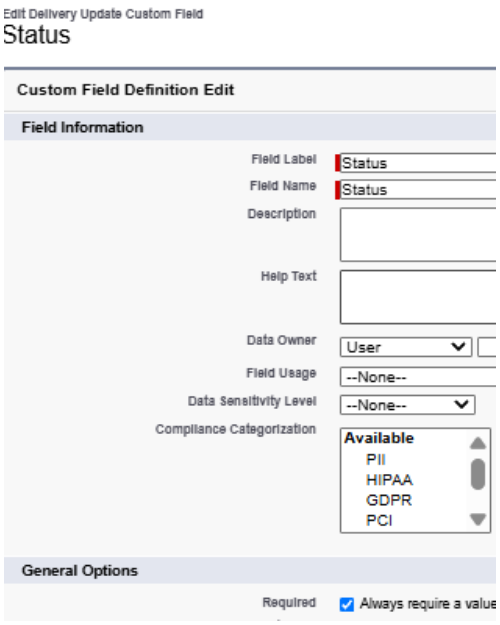
Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes

Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	Date	Yes
Remarks	Remarks__c	Text	No

Status Picklist Values:

- In Transit
- Out for Delivery
- Delivered
- Delayed

Required → Save



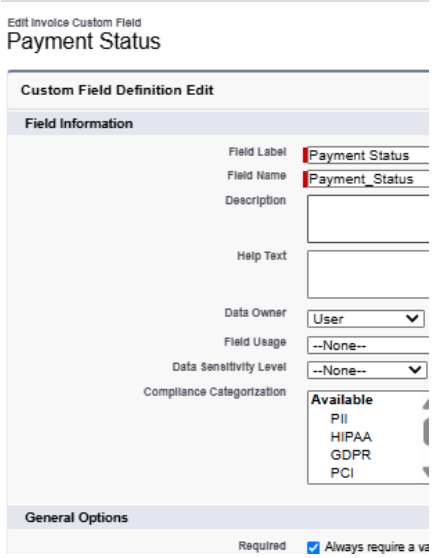
Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status Values

- Paid
- Unpaid
- Pending

Required → Save



Relationships :

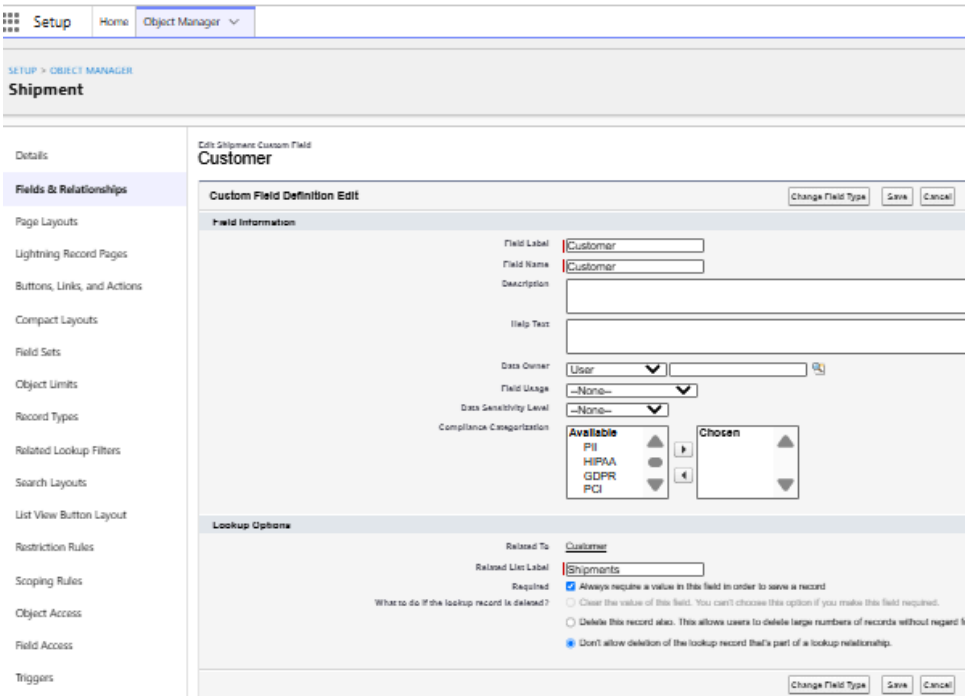
Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery_Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Relationship Creation Steps :

Activity 1 – Shipment → Customer

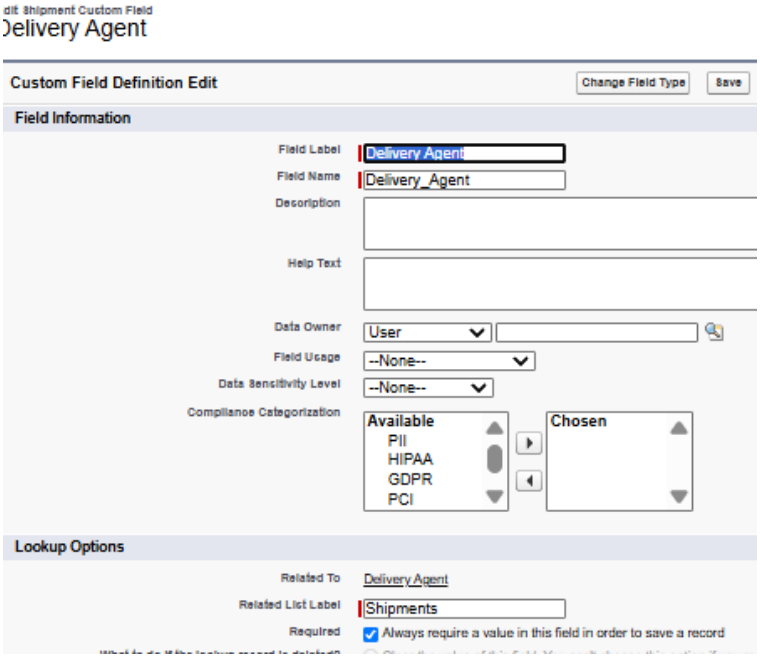
- Setup → Shipment__c → Fields → New

- Lookup Relationship
- Related Object: **Customer__c**
- Field Label: **Customer**
- Save



Activity 2 – Shipment → Delivery Agent

- Setup → Shipment__c
- New → Lookup
- Object: **Delivery_Agent__c**
- Label: **Assigned Agent**
- Save



Activity 3 – Delivery Agent → Branch

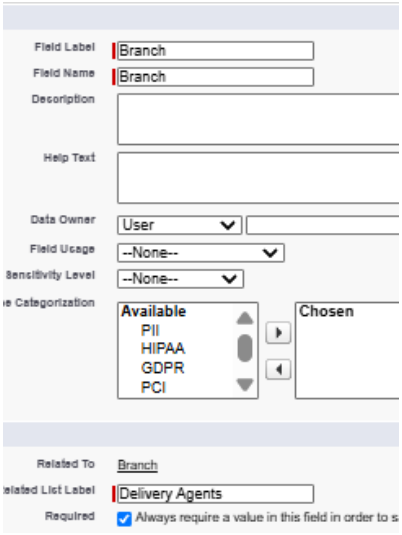
- Setup → Delivery_Agent__c

- New → Lookup

- Related Object: **Branch__c**

- Label: **Branch**

- Save



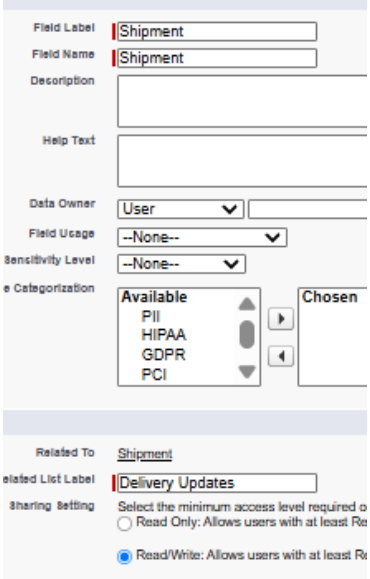
Activity 4 – Delivery Update → Shipment

- Setup → Delivery_Update__c

- New → **Master-Detail**

- Related Object: **Shipment__c**

- Save



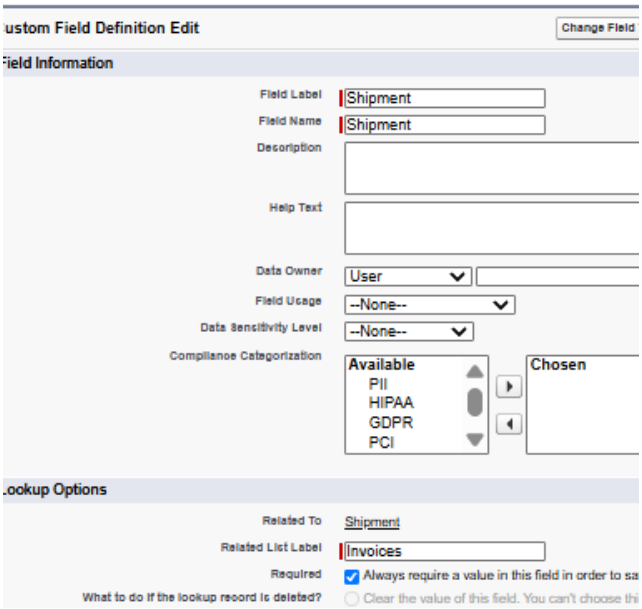
Activity 5 – Invoice → Shipment

- Setup → Invoice__c

- New → Lookup

- Object: **Shipment__c**

- Save



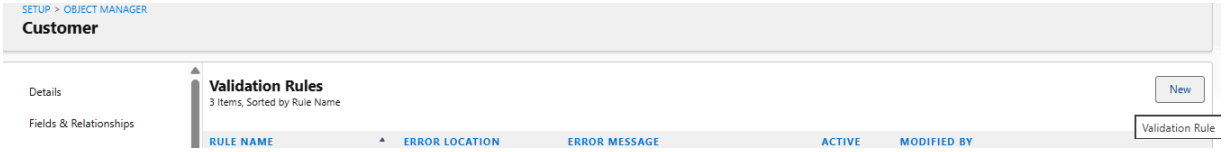
Milestone 5-Validation Rules:

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1:Create Validation Rules on Customer_c

Validation Rule: Email Must Not Be Blank Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New

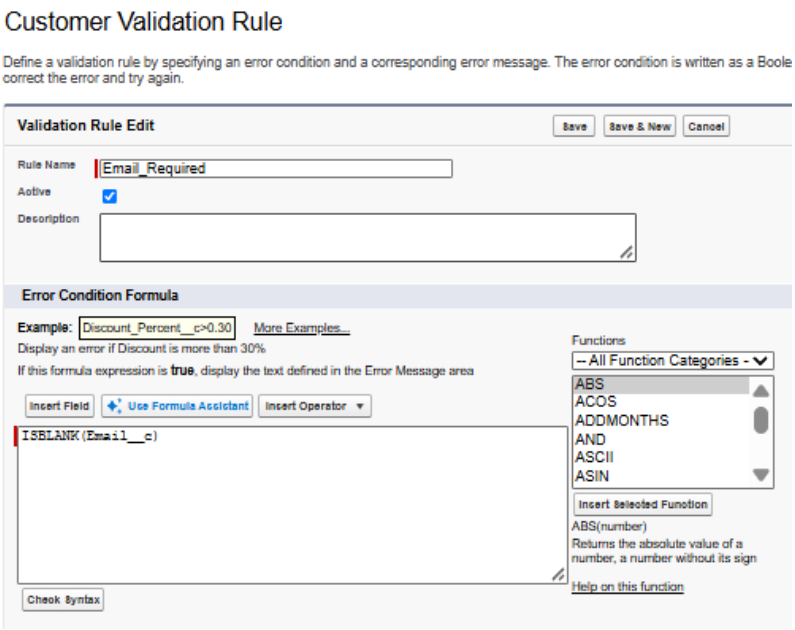


- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.
- Error Location: Field → Email
- Click Save.



2. Validation Rule: Phone Must Be 10 Digits

Steps:

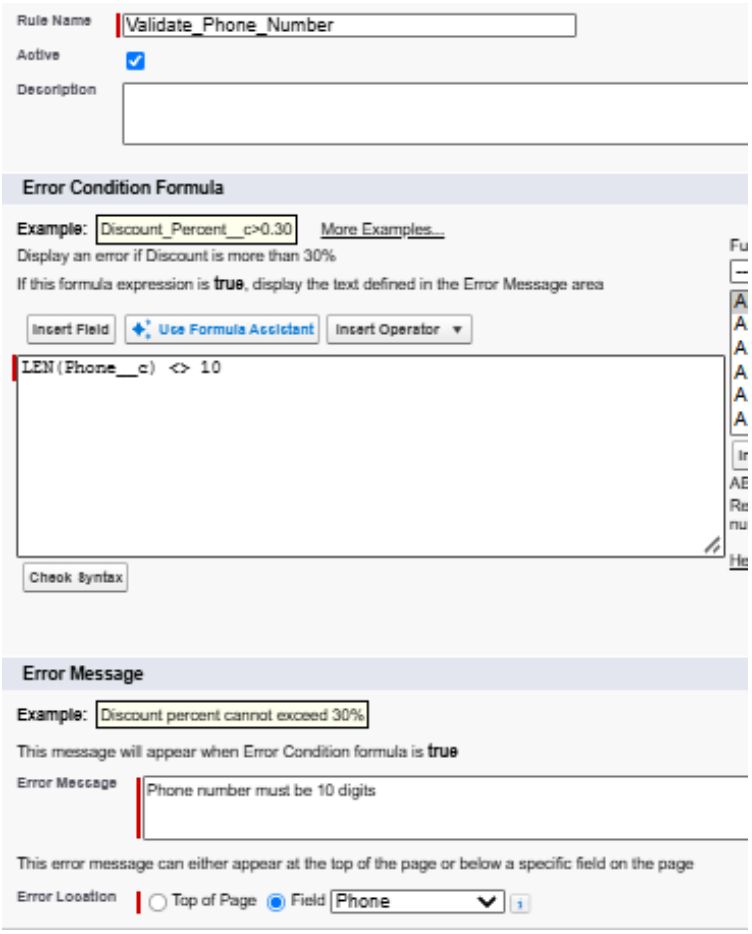
- Setup → Object Manager → Customer__c

- Validation Rules → New
- Rule Name: Validate_Phone_Number
- Check Active

Formula:

LEN(Phone__c) <> 10

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save



3. Validation Rule: Pincode Must Be 6 Digits

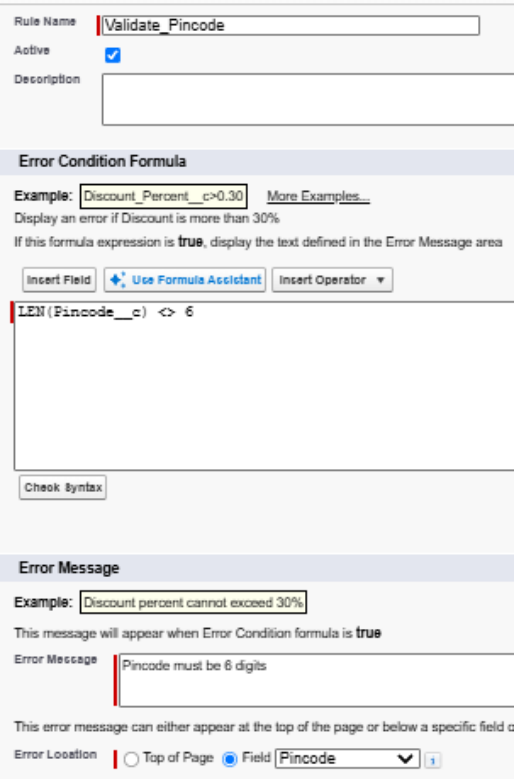
Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode
- Active → Yes

Formula:

LEN(Pincode__c) <> 6

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode
- Save



Activity 2:Create Validation Rules on Shipment_c

1. Validation Rule: Weight Must Be Greater Than Zero

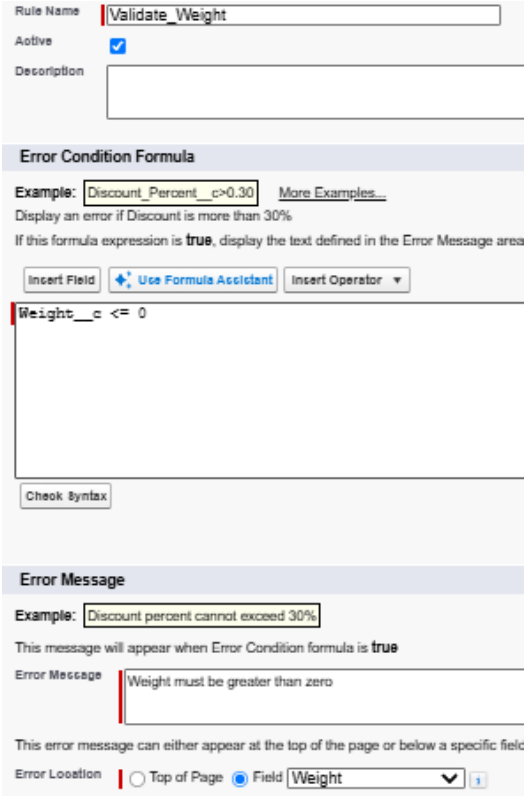
Steps:

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

Weight__c <= 0

- Check Syntax
- Error Message: **Weight must be greater than zero.**
- Location: Field → Weight
- Save



2. Validation Rule: Price Must Be Greater Than Zero

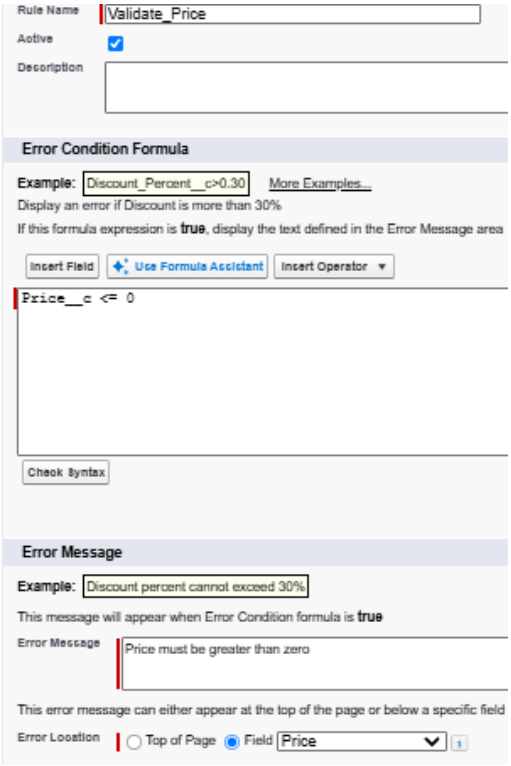
Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

Price__c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save



3. Validation Rule: Expected Date Cannot Be in the Past

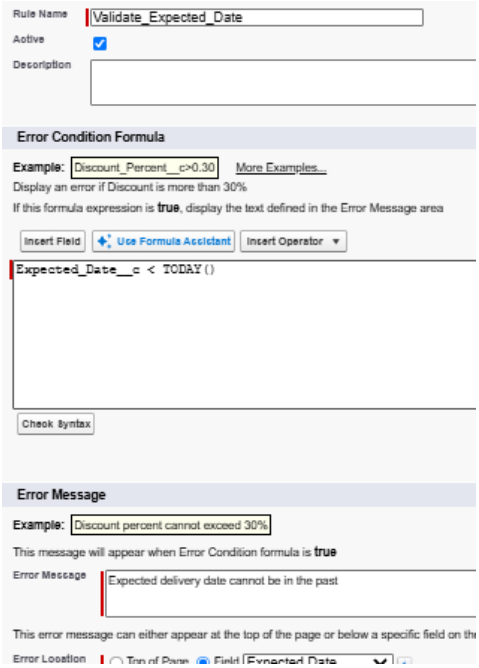
Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Expected_Date

Formula:

Expected_Date__c < TODAY()

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**
- Location: Field → Expected Date
- Save



Activity 3:Create Validation Rules on Delivery_Agent_c

1. Validation Rule: Email Must Be Valid

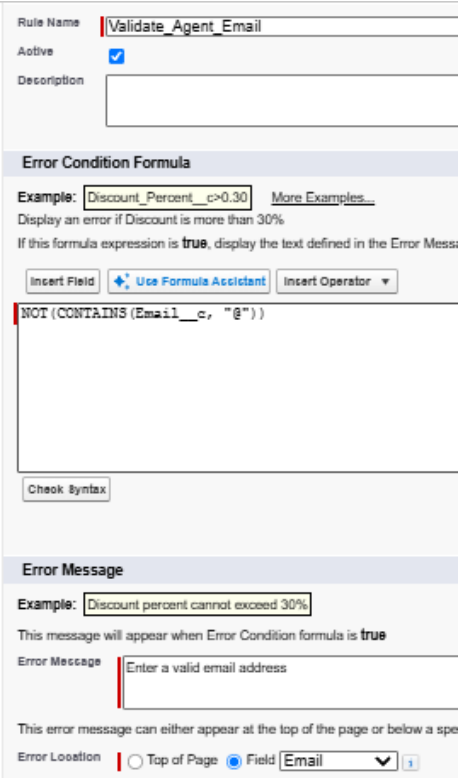
Steps:

- Setup → Object Manager → Delivery_Agent__c
- Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

Formula:

NOT(CONTAINS(Email__c, "@"))

- Error Message: **Enter a valid email address.**
- Location: Field → Email
- Save



Activity 4:Create Validation Rules on Branch_c

1. Validation Rule: Branch Code Must Not Be Blank

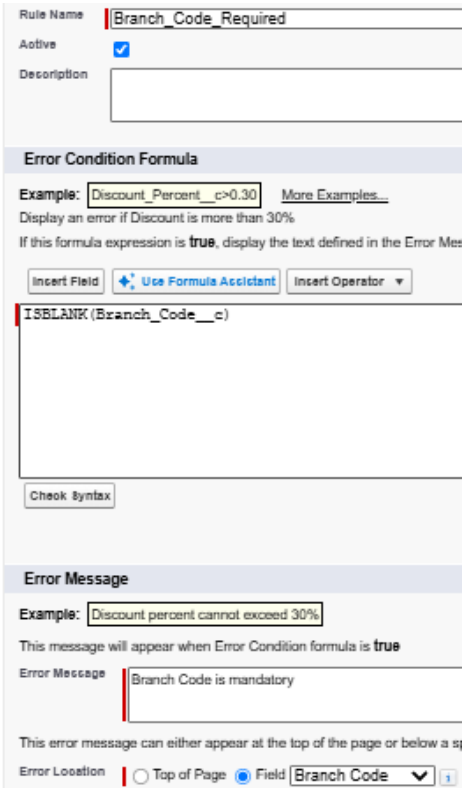
Steps:

- Setup → Object Manager → Branch__c
- Validation Rules → New
- Rule Name: Branch_Code_Required

Formula:

ISBLANK(Branch_Code__c)

- Error Message: **Branch Code is mandatory.**
- Location: Field → Branch Code
- Save



Activity 5:Create Validation Rules on Delivery_Update_c

1. Validation Rule: Remarks Required If Status is Delayed

Steps:

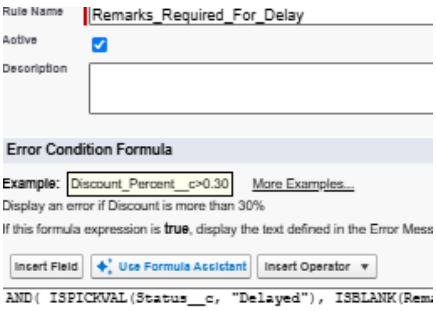
- Setup → Object Manager → Delivery_Update__c
- Validation Rules → New
- Rule Name: Remarks_Required_For_Delay

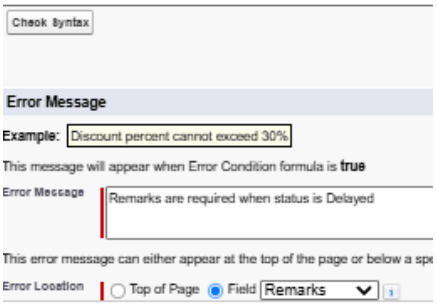
Formula:

```
AND(
 ISPICKVAL(Status__c, "Delayed"),
 ISBLANK(Remarks__c)
)
```

- Error Message: **Remarks are required when status is Delayed.**

- Location: Field → Remarks
- Save





Milestone 6 - Page Layouts:

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.

They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Activity 1: Edit Page Layout for customer_c

Steps:

- Setup → Object Manager → Customer__c
- Click Page Layouts

- Click the default layout (usually Customer Layout)

Sections to Add & Arrange:

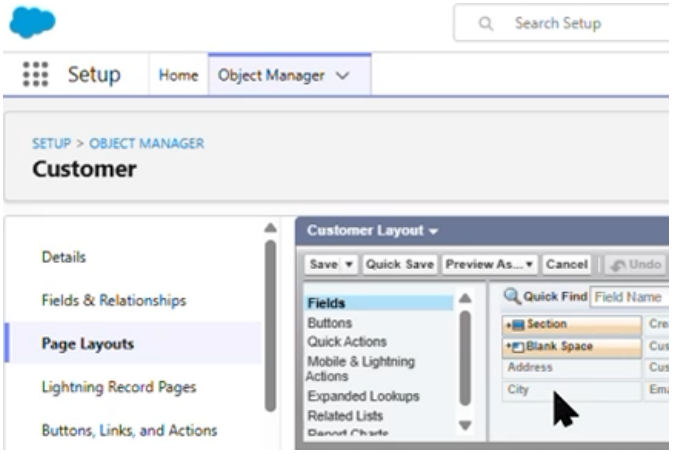
- Open the default layout
- In the top palette, click “Section”
- Drag the Section and drop it anywhere on the layout

A pop-up appears → Fill details:

- Section Name: “Customer Information”
- Layout:
    - 1-Column or
    - 2-Column (recommended)
- Tab-Key Order: Top-Down
- Click OK

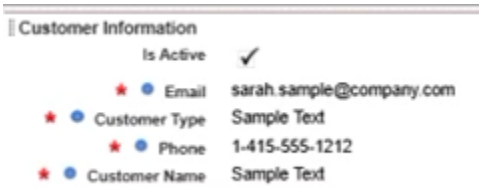
The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.



Section 1: Customer Information

- Customer Name
- Customer Type
- Email
- Phone
- Is Active



Section 2: Address Details

- Address
- City
- State
- Pincode



9. Edit the layout as needed

10. Click Save.

Activity 2: Edit Page Layout for Shipment_c

Steps:

- Setup → Object Manager → Shipment__c
- Click Page Layouts
- Open the default Shipment Layout

Sections to Add:

Section 1: Shipment Details

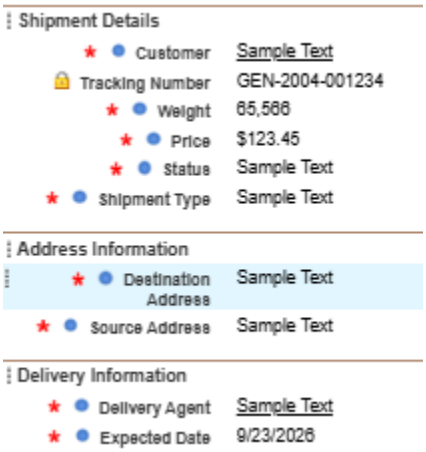
- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Section 2: Address Information

- Source Address
- Destination Address

Section 3: Delivery Information

- Expected Date
- Assigned Agent
- Arrange fields properly
- Click Save



Activity 3: Edit Page Layout for Delivery_Agent_c

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Page Layouts
- Open default Delivery Agent Layout

Arrange Fields:

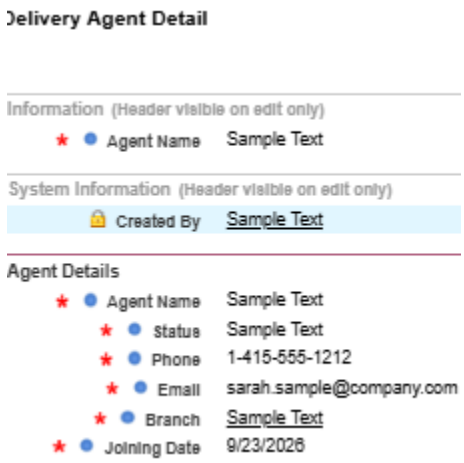
Agent Details

- Agent Name
- Email
- Phone
- Branch

- Status
- Joining Date

Additional Information

- Related Shipments
- Related Delivery Updates
- Edit → Save



Activity 4: Edit Page Layout for Branch_c

Steps:

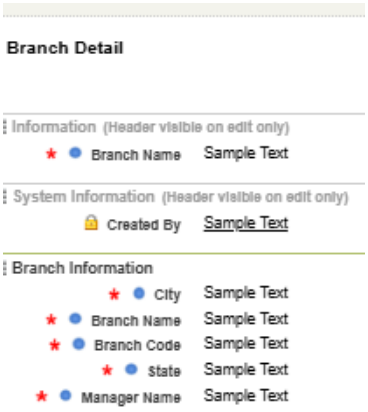
- Setup → Object Manager → Branch__c
- Page Layouts
- Open default Branch Layout

Sections:

Branch Information

- Branch Name
- Branch Code

- City
- State
- Manager Name
- Arrange fields
- Click Save



Activity 5: Edit Page Layout for Delivery_Update_c

Steps:

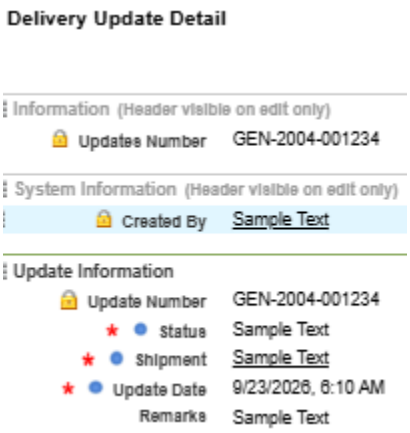
- Setup → Object Manager → Delivery_Update__c
- Page Layouts
- Open default Delivery Update Layout

Sections:

Update Information

- Update Number
- Shipment
- Status

- Update Date
- Remarks
- Arrange properly
- Click **Save**



Activity 6: Edit Page Layout for Invoice_c

Steps:

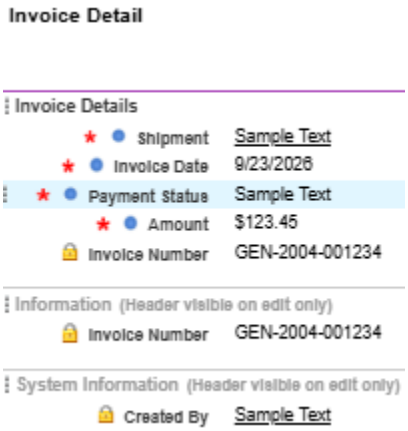
- Setup → Object Manager → Invoice__c
- Page Layouts
- Open default Invoice Layout

Sections:

Invoice Details

- Invoice Number
- Shipment
- Amount
- Invoice Date

- Payment Status
- Arrange fields as needed
- Click Save



Milestone 7 - Flows:

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

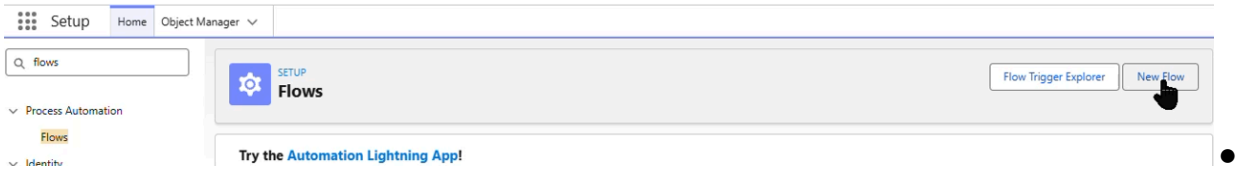
Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

Part 1: Create the Record-Triggered Flow

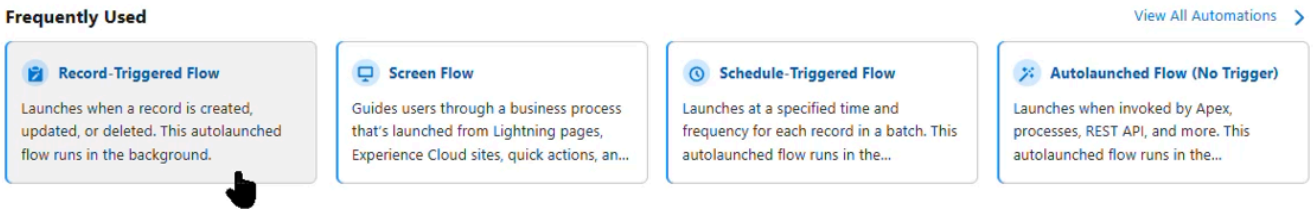
Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button



Select Record-Triggered Flow

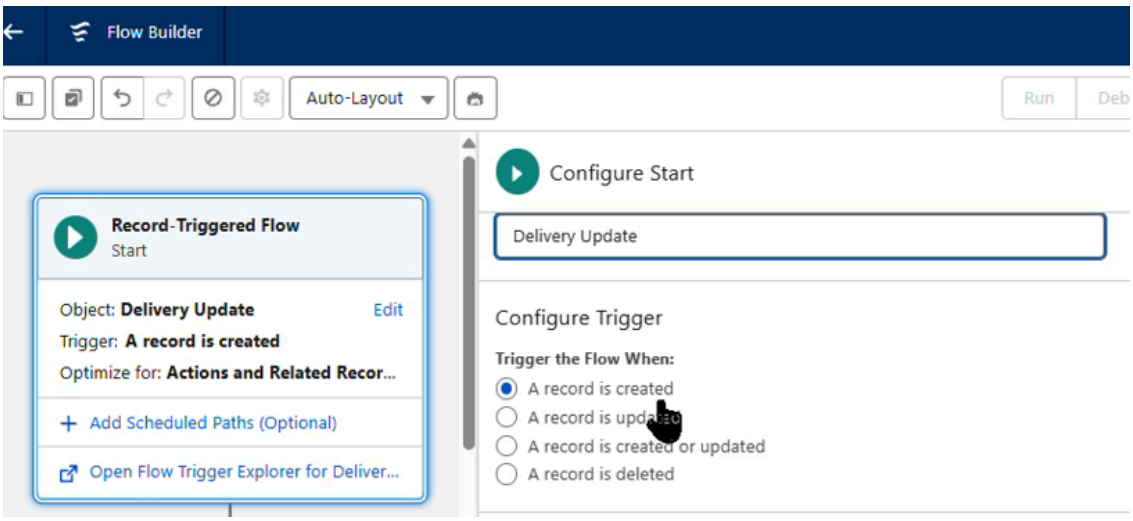
- Click Create



Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
    - Select A record is created
- Condition Requirements:
    - Choose: None (Always Run the Flow)
- Optimize the Flow For:
    - Select Actions and Related Records
- Click Done

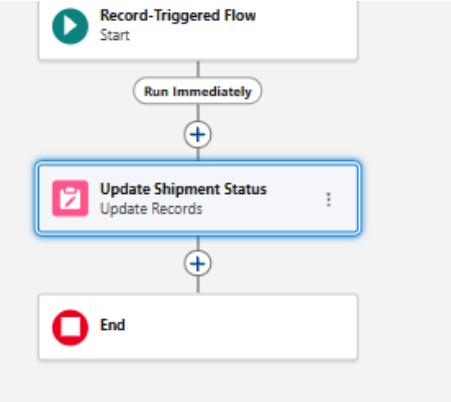


Part 2: Update Related Shipment Record

Step 3: Add “Update Records” Element

- Click the + icon under the Start element

- Select Update Records

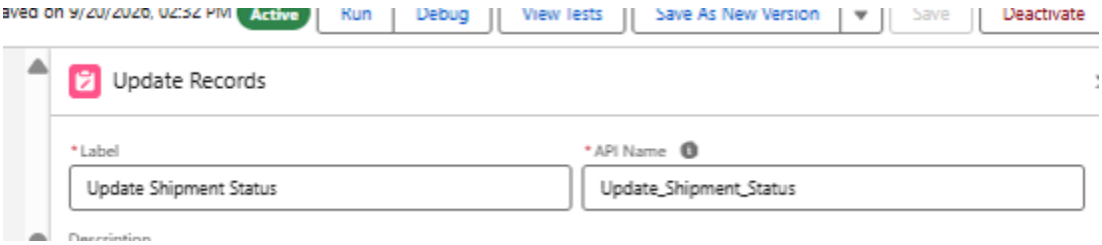


Configure the Update Records Element:

- Label: Update_Shipment_Status

- How to Find Records to Update:

    - Select Specify conditions to identify records



Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals
- Value: {!$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

Set Field Values to Update

Add field mapping:

Shipment Field	Value Source
Status__c	{!$Record.Status__c}

(This copies status from Delivery Update to Shipment)

- Click Done

Part 3: Save and Activate

Step 4: Save Flow

- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

Step 5: Activate Flow

- Click Activate
- Flow status should show Active

Milestone 8 - The Lightning App:

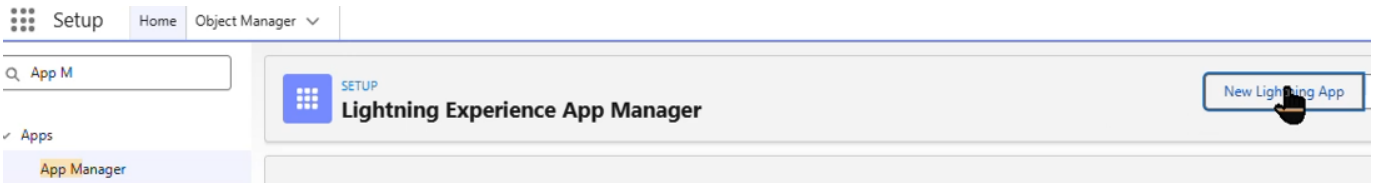
What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Creating a Lightning App

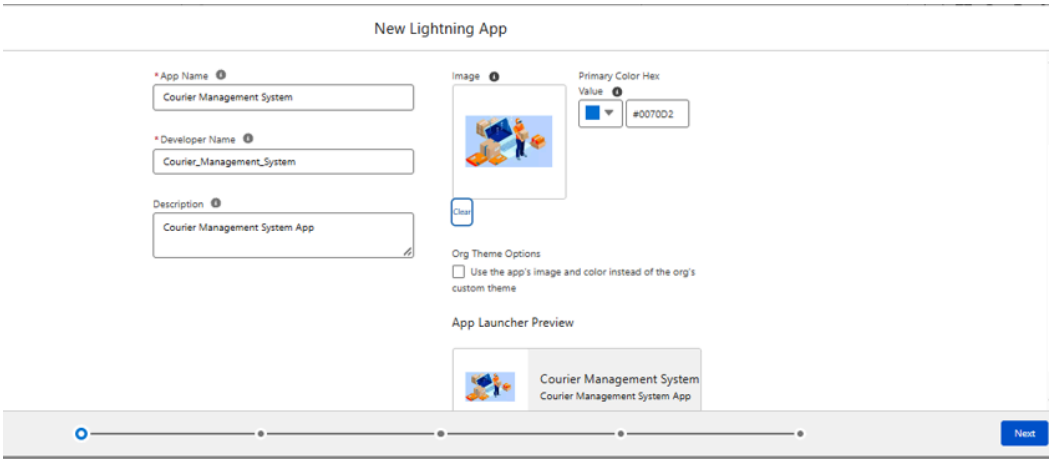
Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.



   ○ Enter the Courier Management System as the App Name.

3. Upload an image or icon for the app, then click Next.
4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.



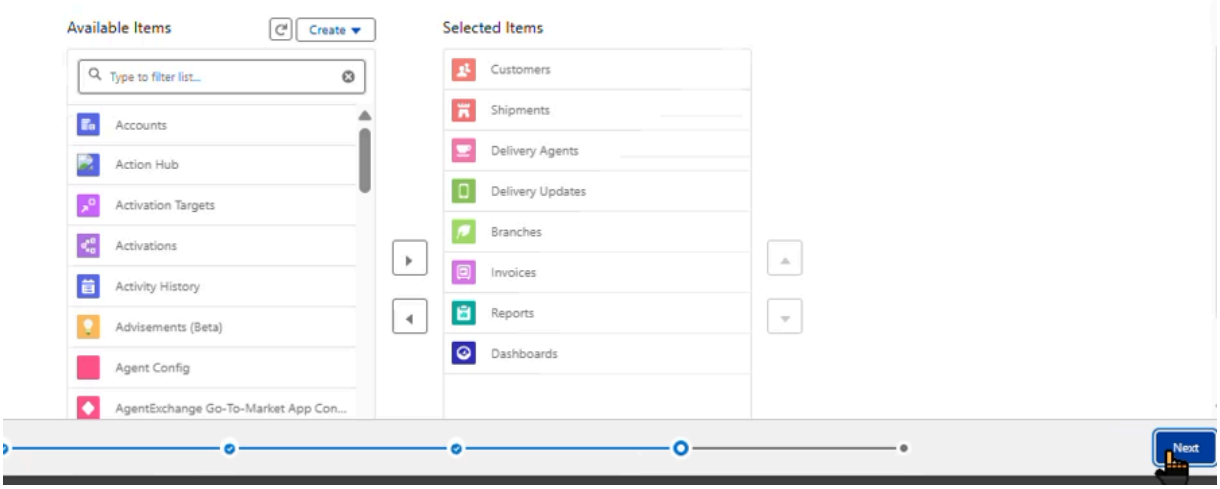
6. From Available Items, select the following and move them to Selected Items:

   ○ Customer

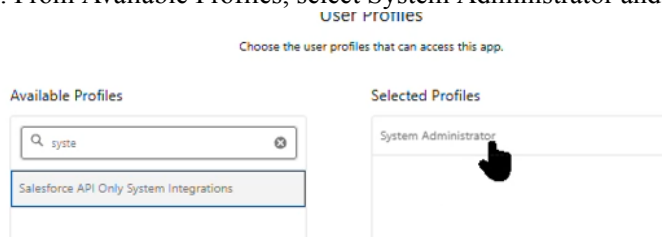
   ○ Shipment

- Delivery Agent
- Branch
- Delivery Update
- Invoice
- Reports
- Dashboards

7. Click Next.



8. From Available Profiles, select System Administrator and move it to Selected Profiles.



9. Click Save & Finish.



Milestone 9 - Profile:

What is a profile?

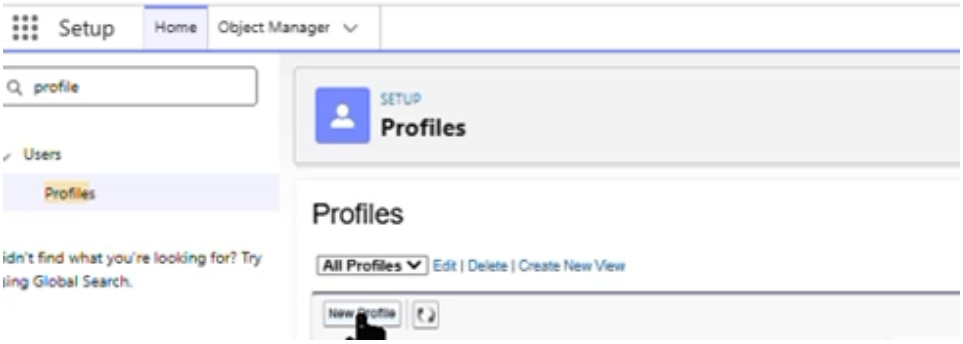
A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but users can be assigned a single profile at a time.

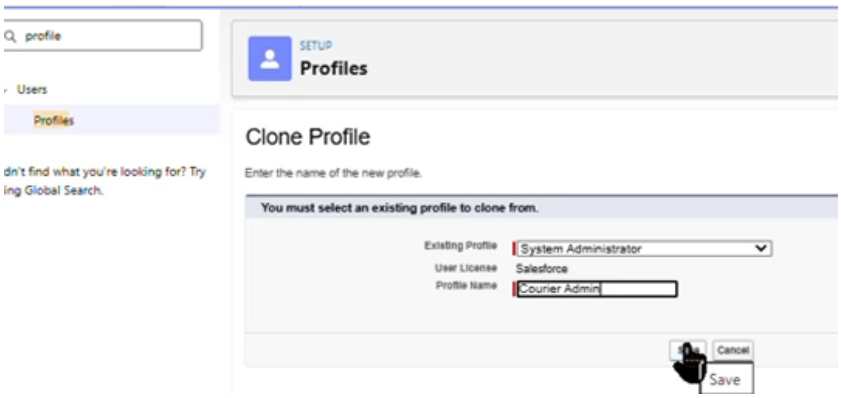
Activity 1: create a Custom Profile - Courier Admin

Create a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.



4. Select Existing Profile as: System Administrator.
5. Enter Profile Name: Courier Admin.
6. Click Save.



Why use System Administrator as Base?

It already has:

- Modify All Data
- Customize Application
- Full CRUD access on all objects

By cloning it, the **Courier Admin profile automatically inherits full access**.
You can later restrict permissions if required.

Activity 2: Create a Custom Profile - Branch Manager

Steps:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.
4. Select Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.
6. Click Save.
7. Click Edit on the Branch Manager profile.

Set App & Tab Settings

Under Assigned Apps / Custom App Settings, set the default app to:

Courier Management System App

Ensure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

Object	Permissions to Grant
Customer__c	Create, Read, Edit

Shipment__c	Create, Read, Edit
Delivery_Agent__c	Read, Edit

Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read

Session & Password Settings

Under Session Settings:

- Session Timeout: 2 Hours of Inactivity

Under Password Policies:

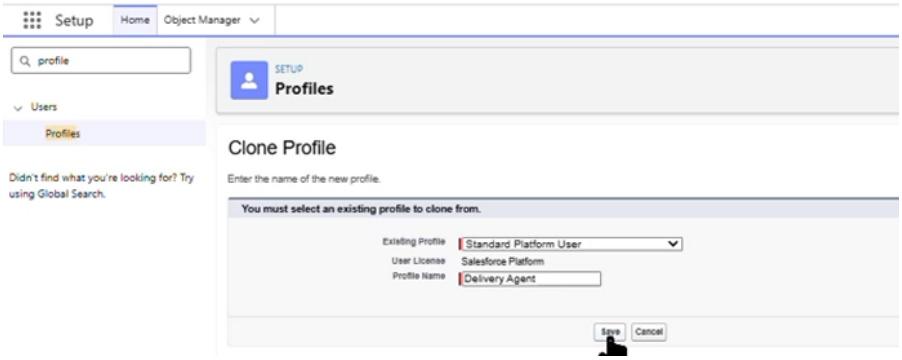
- User passwords expire in: Never Expires
- Minimum Password Length: 8

Click Save.

Activity 3: Create a custom Profile - Delivery Agent

Steps:

1. Setup → Quick Find → Profiles
2. Click New Profile
3. Existing Profile: Standard Platform User
4. Profile Name: Delivery Agent
5. Click Save



App & Tab Access

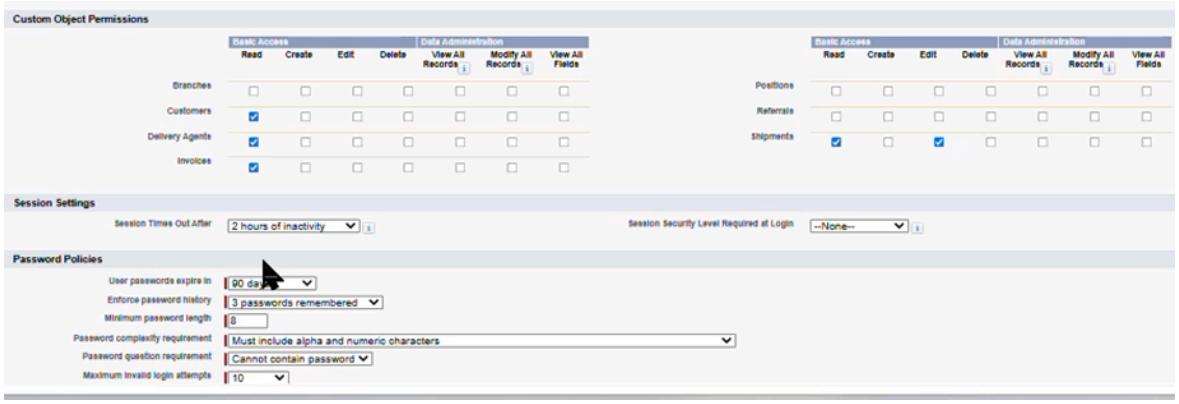
Default App: Courier Management System

Visible Tabs:

- Shipments
- Delivery Updates

Object Permissions :

Object	Permissions
Shipment__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Customer__c	Read
Invoice__c	Read



Restrictions

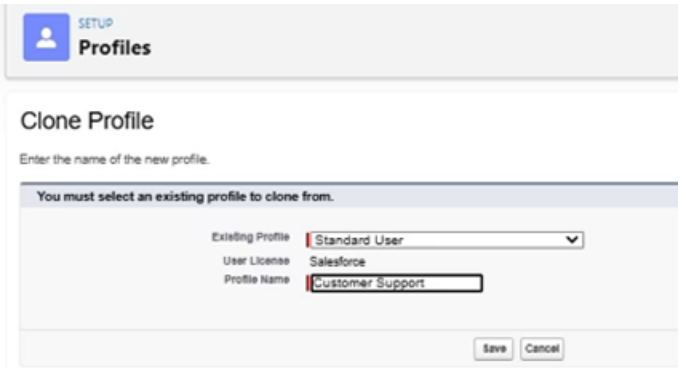
- No access to Branch editing
- No delete permissions
- Limited reporting access

Click Save.

Activity 4: Create a Custom Profile - customer Support

Steps:

- Setup → Profiles → New Profile
- Base Profile: Standard User
- Profile Name: Customer Support



Access Configuration

Tabs Visible:

- Customers
- Shipments
- Delivery Updates

Object Permissions :

Object	Access
Customer__c	Read, Edit
Shipment__c	Read, Edit
Delivery_Update__c	Read

Invoice__c Read

Session Settings

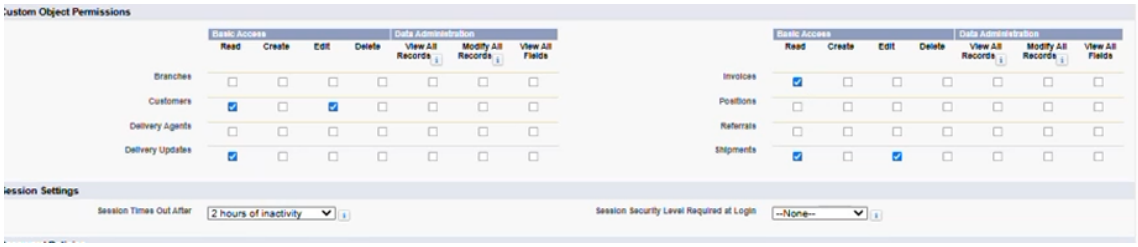
- Session Timeout: 2 Hours
- Password Length: 8

Click Save

Final Profiles Created in Project :

Profile Name	Purpose

Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking



Milestone 10 - Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of courier Admin Role

Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.

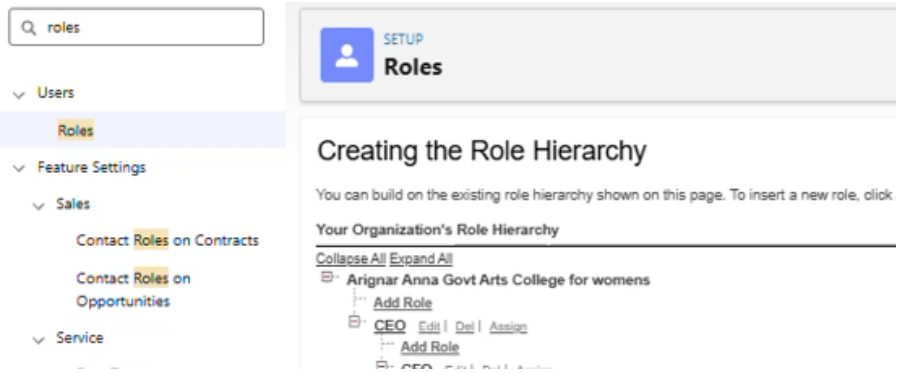
2. Select Set Up Roles.

3. Click on Expand All and click Add Role under CEO.

4. Give Label as “Courier Admin”.

    ○ The Role Name will auto-populate.

5. Click Save.



Activity 2: Creating Branch Manager Role Under courier admin Role

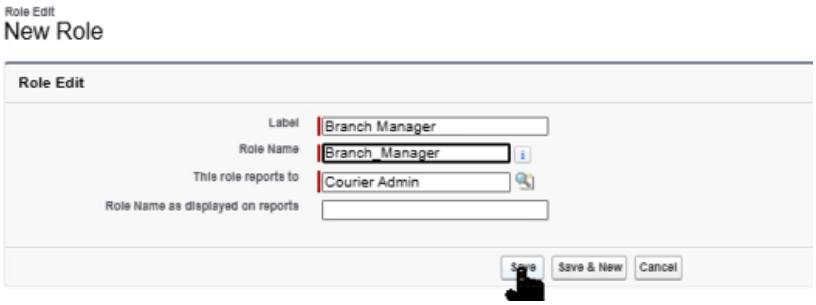
Create another role:

1. Create a role with the name “Branch Manager”.

2. Below the Courier Admin, click on Add Role.

3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.

4. Click Save.



Activity 3: Creating Delivery Agent Role Under Branch Manager role

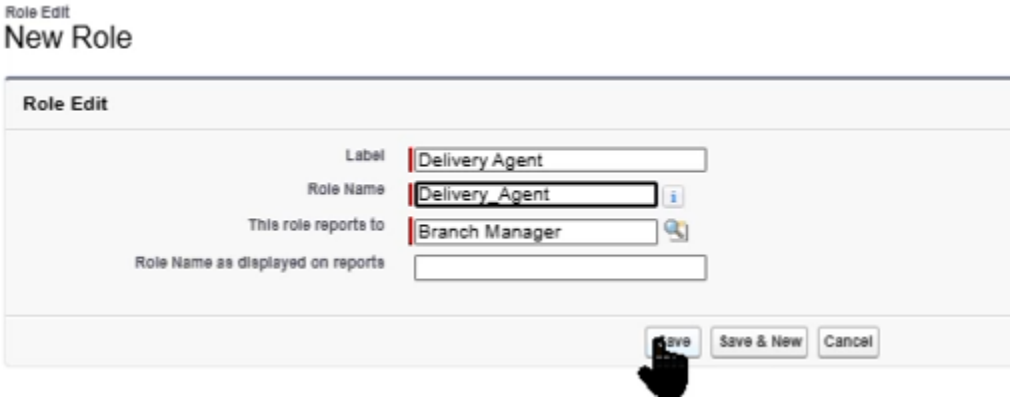
Steps:

1. Below the Branch Manager role, click on Add Role.

2. Enter Label as “Delivery Agent”.

    ○ Role Name will auto-populate.

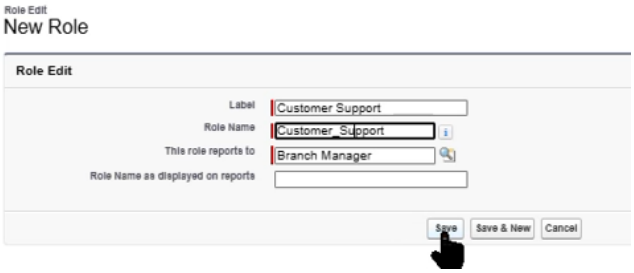
3. Click Save.



Activity 4: Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.



Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking This hierarchy ensures proper data sharing and security in the Courier Management System.

Milestone 11 - User:

What is a user?

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

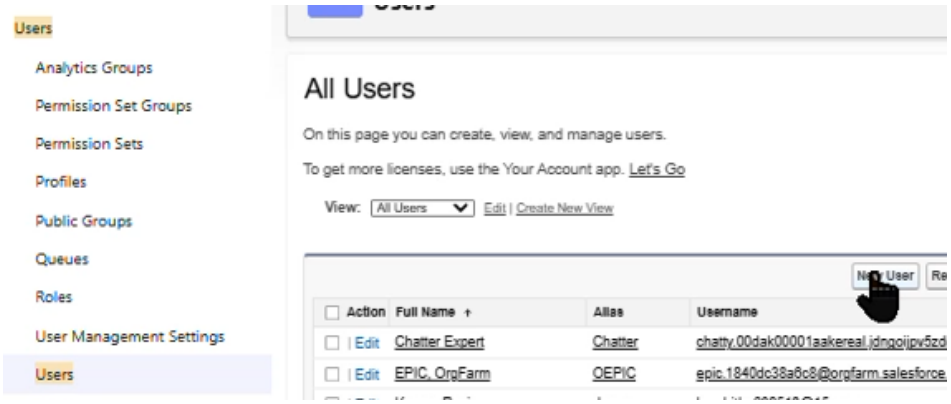
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.

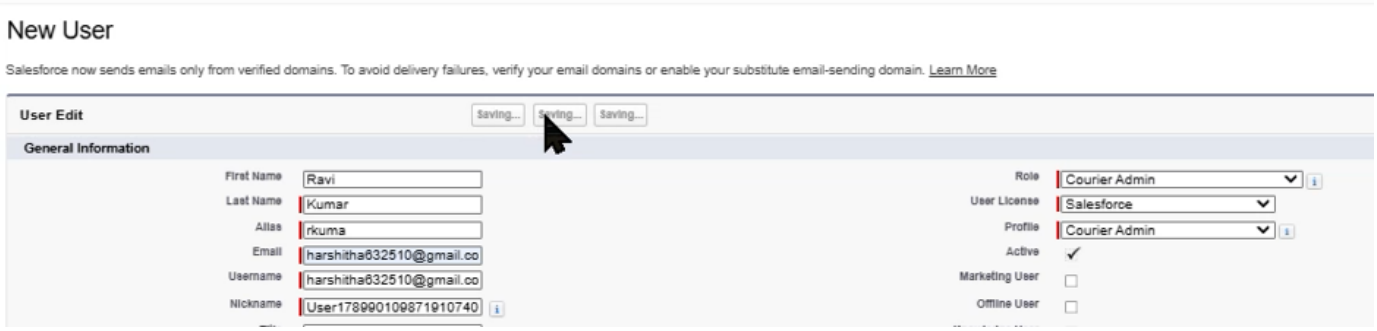
2. Click New User.



Fill in the Fields:

Field Name	Value
First Name	Enter first name (Example: Ravi)
Last Name	Enter last name (Example: Kumar)

Alias	Give an alias name
Email ID	Enter your personal email ID
Username	Must be in the format: text@text.text and must be unique
Nickname	Give a nickname
Role	Select **Courier Admin**
User License	Salesforce
Profile	**Courier Admin**



3. Click Save.

Activity 2: Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform I	Branch Manager

		Agent	
Customer Support	Customer Support	Salesforce Platform C Support	Branch Manager

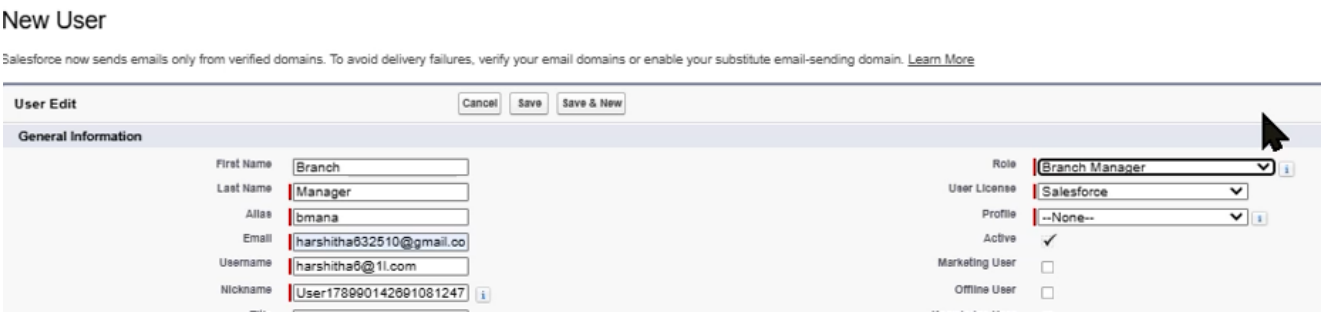
Example:

To create a Branch Manager User:

- Go to Setup → Users → New User
- Fill details like:

Field	Example Value
First Name	Arjun
Last Name	Reddy
Email	arjun@gmail.com
Username	arjun.manager@cms.com
Role	Branch Manager
Profile	Branch Manager

- Click Save

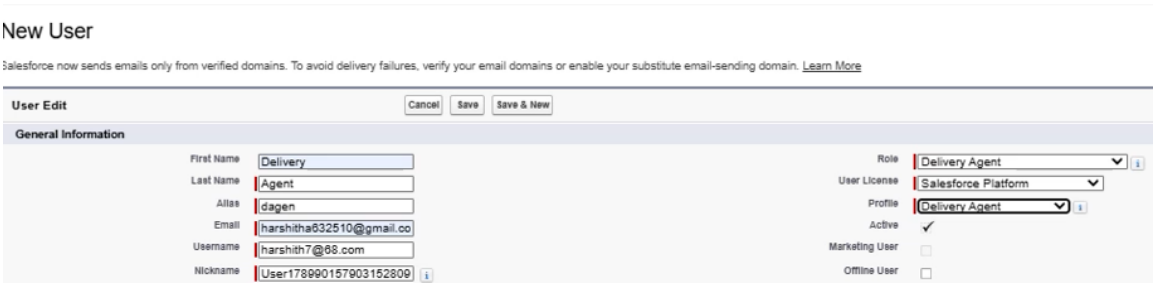


\

To Create Delivery Agent User:

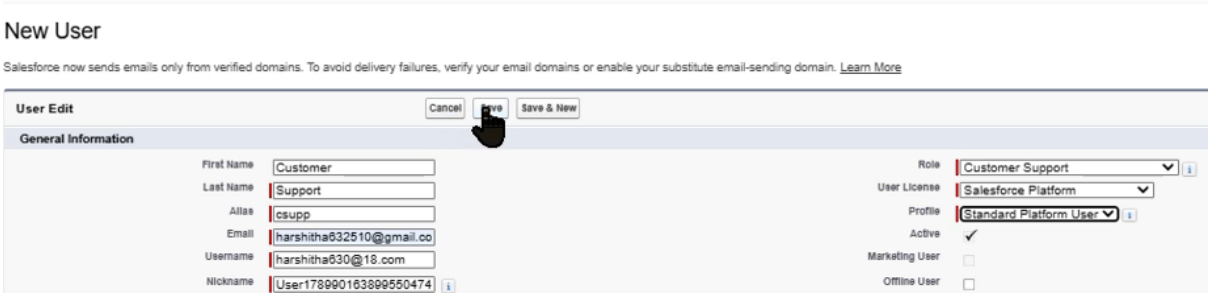
Follow same steps with:

- Role: Delivery Agent
- Profile: Delivery Agent



To Create Customer Support User:

- Role: Customer Support
- Profile: Customer Support



Important Points

- Every user must have:

- Unique username
- Valid email
- Assigned Role
- Assigned Profile

- Profiles control permissions
- Roles control data visibility

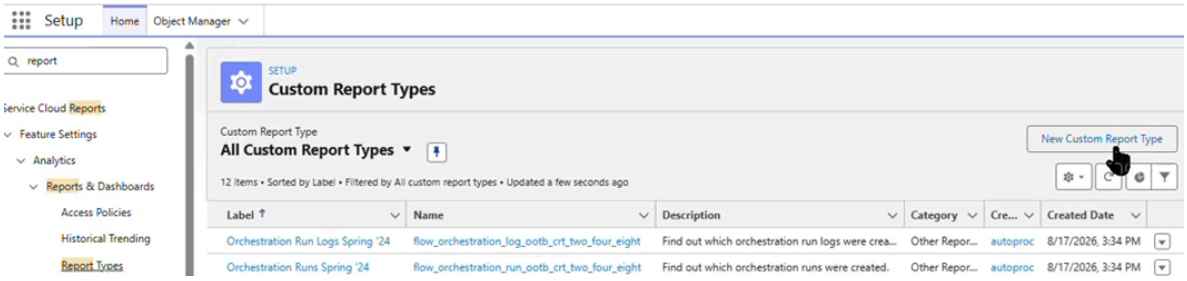
Milestone 12 - Reports & Dashboards:

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

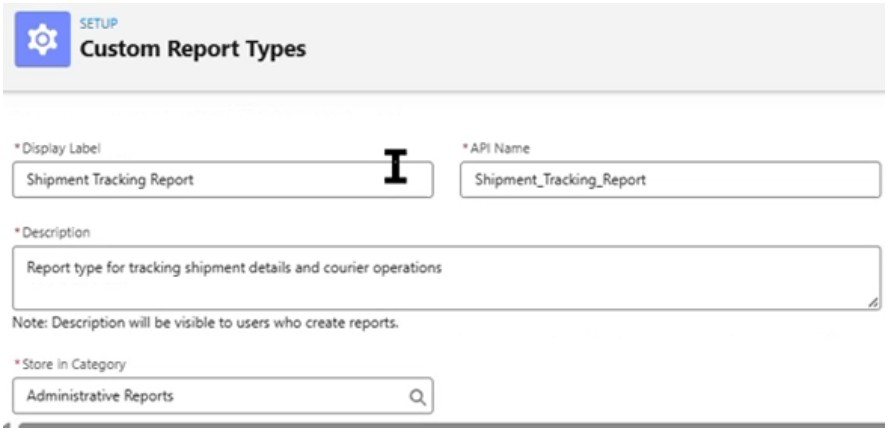
1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.



Step 2: Create Shipment Report Type

- Primary Object: Shipment__c
- Label: Shipment Tracking Report

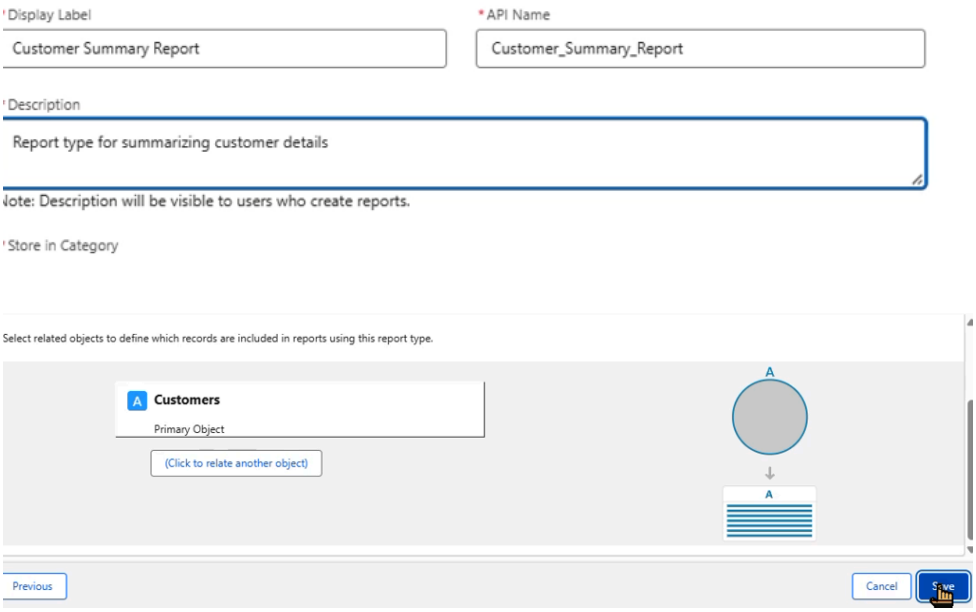
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save





Step 3: Create Customer Report Type

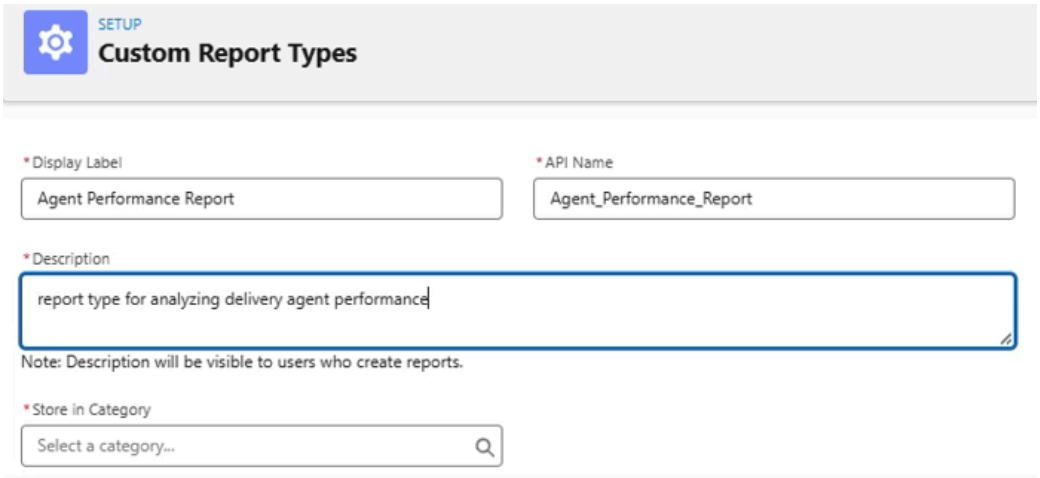
- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

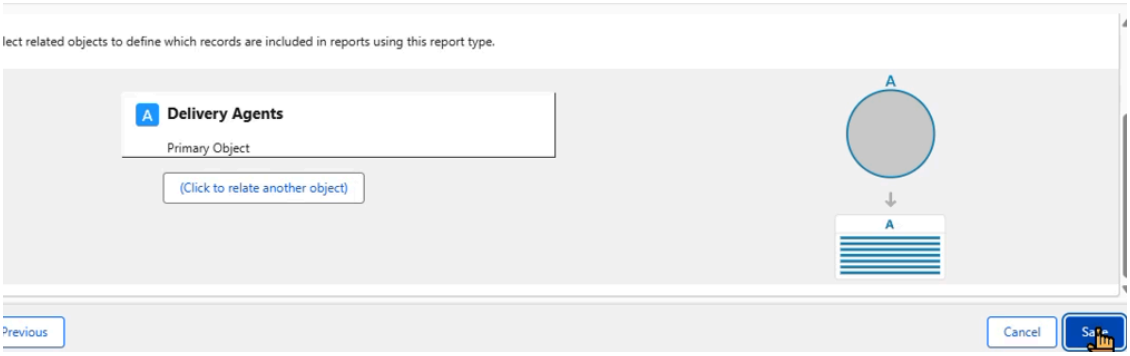


Step 4: Create Agent Performance Report Type

- Primary Object: Delivery_Agent__c

- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save



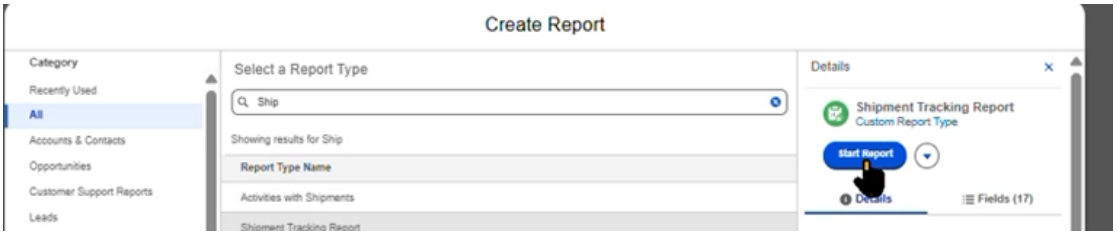


Activity 2: Create Reports

Report 1: Shipment Status Summary Report

Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue



Step 2: Add Columns

Add these fields:

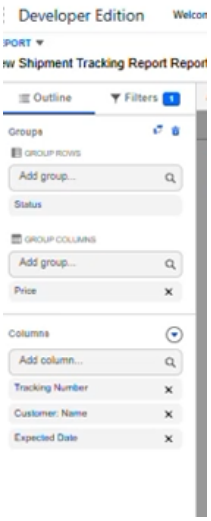
- Tracking Number
- Customer
- Status
- Expected Date
- Assigned Agent

Step 3: Add Filters

- Show Me: All Shipments
- Date Filter: This Month
- Status Filter: Select specific Status (Optional)

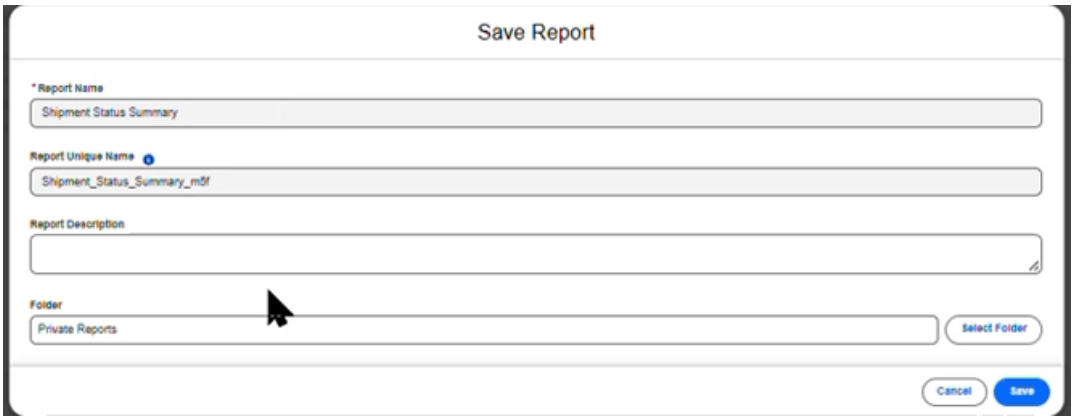
Step 4: Group & Summarize

- Group rows by Status__c
- Add summary on Price



Step 5: Save Report

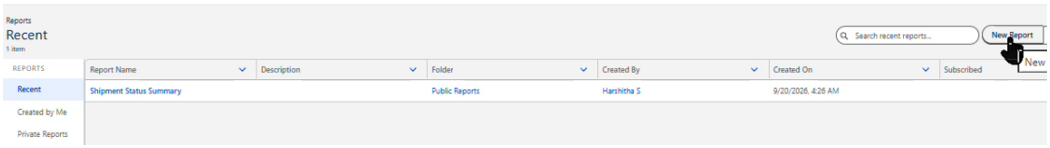
- Click Save & Run
- Name: Shipment Status Summary
- Folder: CMS Reports



Report 2: Agent Performance Report

Step 1: Create Report

- New Report → Choose Agent Performance Report
- Click Continue



Step 2: Add Columns

- Agent Name
- Branch
- Number of Shipments
- Status

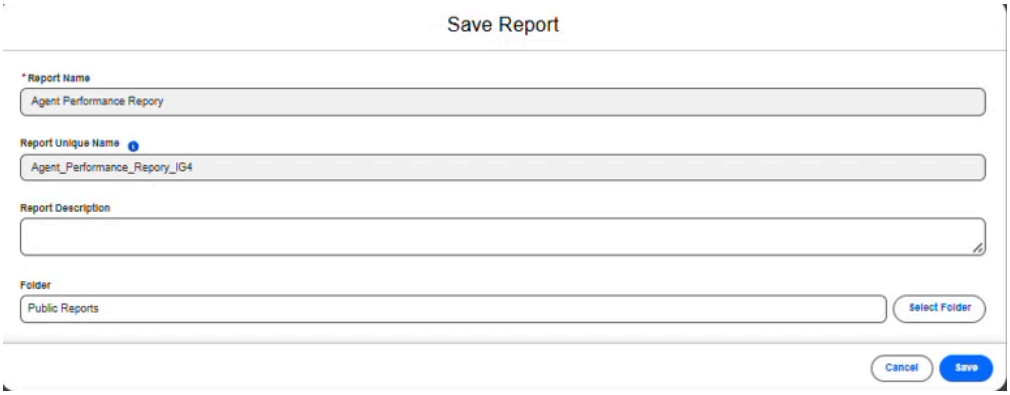
Step 3: Group Data

- Group by Delivery_Agent__c

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports

- Save & Run



Report 3: Revenue Report

Steps

- Create report using Shipment Tracking Report Type

Add fields:

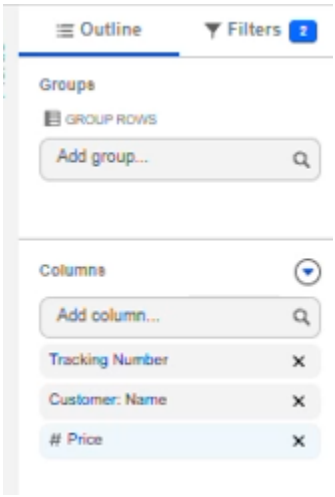
- Tracking Number
- Customer
- Price
- Invoice Status

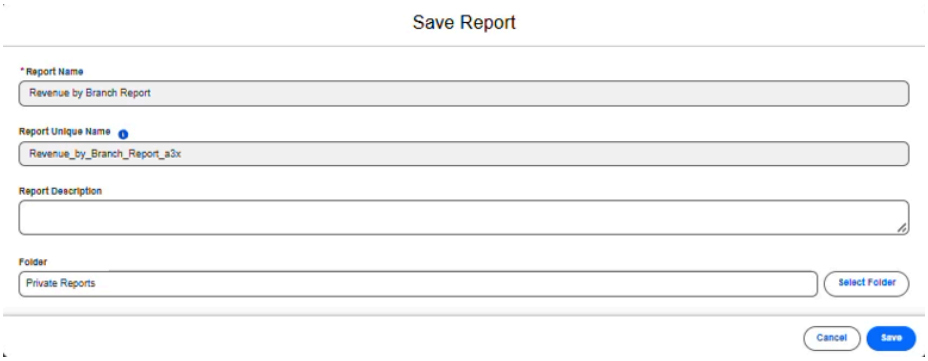
Group by:

- Branch

● Month

Save as: Revenue by Branch Report





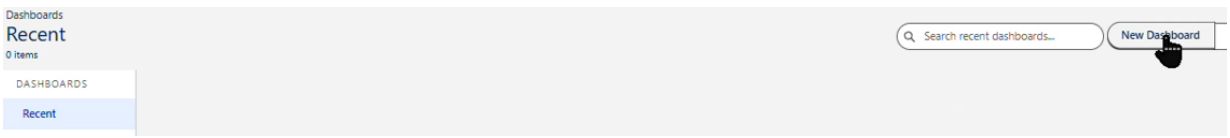
Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)

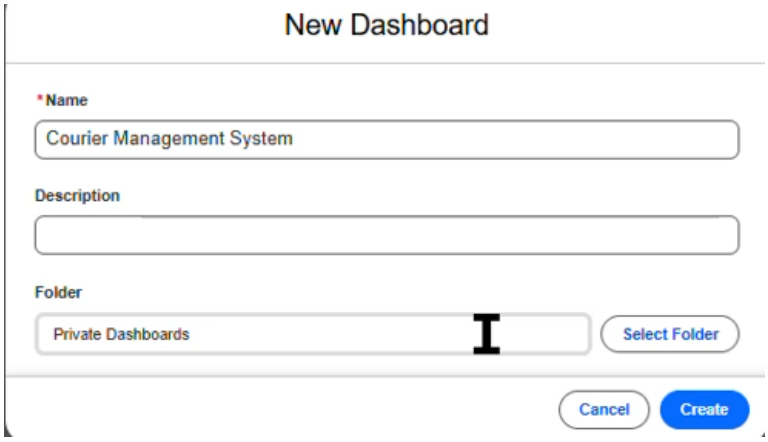
2. Search Dashboards

3. Click New Dashboard

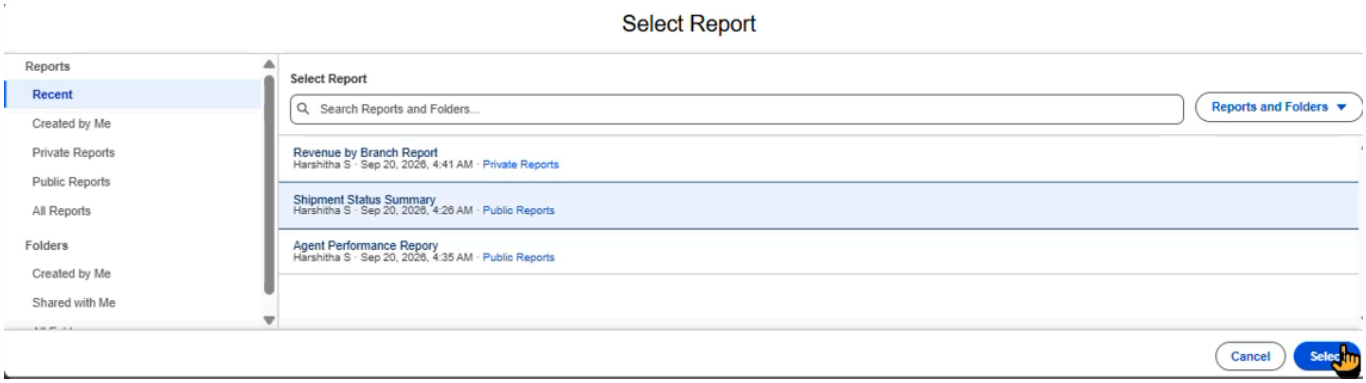


Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create

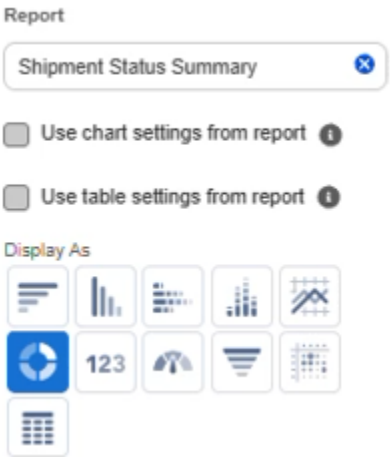


Activity 4: Add Dashboard Components



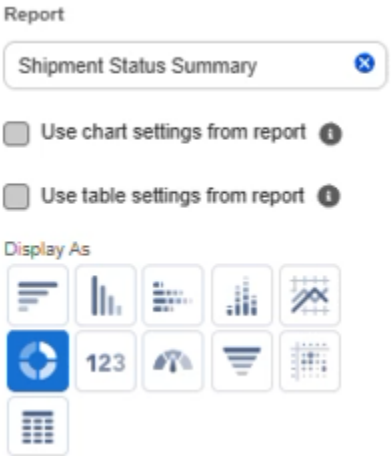
Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add



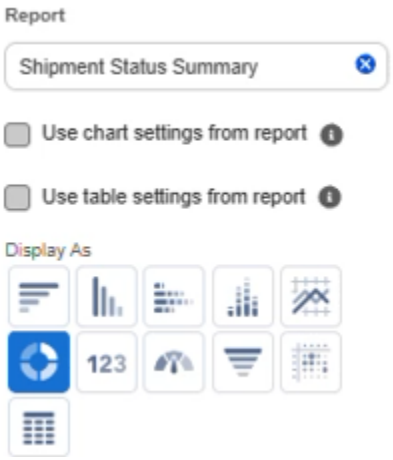
Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add



Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart

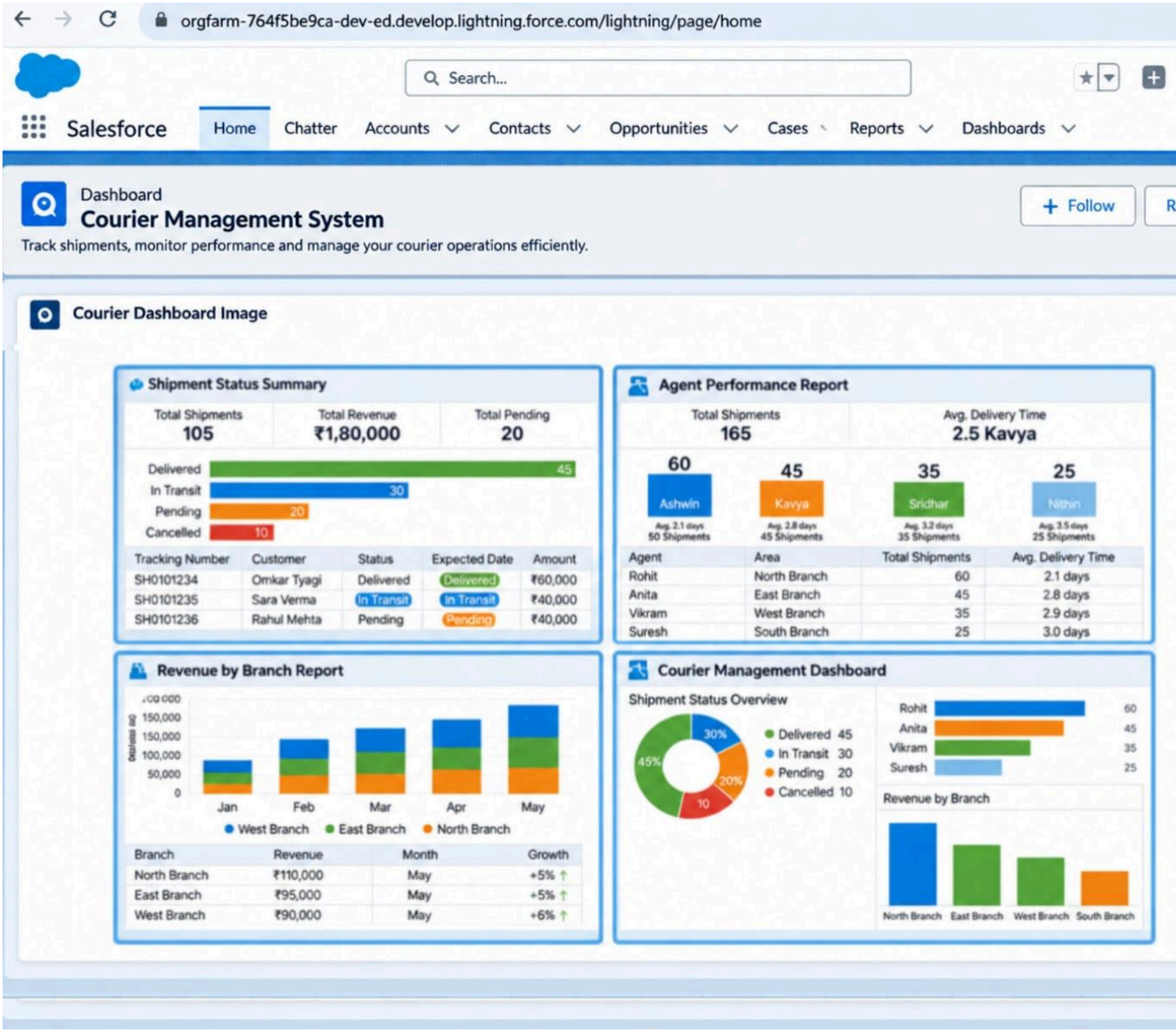


- Click Add

Activity 5:Save & Active dashboard

1. Click Save

2. Click Done



Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

7. FUNCTIONAL & PERFORMANCE TESTING

Test ID	Test Case	Expected Result	Status
TC-01	Create a customer record	Customer record is saved with valid details.	Pass
TC-02	Create shipment without required data	Validation prevents incomplete record.	Pass
TC-03	Assign delivery agent	Shipment shows the selected agent.	Pass
TC-04	Create delivery update	Update is stored against the shipment.	Pass
TC-05	Change delivery status	Shipment status is synchronized by Flow.	Pass
TC-06	Enter invalid weight	Validation message is displayed.	Pass
TC-07	Create invoice	Invoice is linked to the shipment.	Pass
TC-08	Open operational report	Relevant shipment data appears.	Pass
TC-09	Open dashboard	Dashboard components load project metrics.	Pass
TC-10	Check role-based access	User sees only permitted functions and records.	Pass

Performance testing focuses on normal user actions such as opening list views, creating records, saving updates and loading reports. The application should be tested with representative sample data to identify slow configurations, excessive automation or unnecessary page fields.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized courier information
- Traceable shipment lifecycle
- Reduced duplicate manual entry
- Automated status synchronization
- Structured user access
- Operational reports and dashboards
- Cloud-based accessibility
- Expandable data model

Disadvantages

- Requires Salesforce configuration knowledge
- User training is necessary

- Poorly designed automation can increase maintenance effort
- Licensing and platform limits depend on the Salesforce environment
- Internet connectivity is required for normal cloud access

9. Conclusion

The Courier Management System provides a structured Salesforce implementation for handling customer registration, shipment booking, delivery assignment, tracking updates, invoicing and operational reporting. The project demonstrates how custom objects, relationships, validation rules, page layouts, Flow automation, profiles, roles, users, reports and dashboards can be combined into a single courier workflow. By keeping shipment information connected from booking through delivery, the system creates a clearer operational record and a stronger foundation for future enhancements.

10. FUTURE SCOPE

- Customer self-service tracking portal
- SMS and email notification integration
- Mobile-oriented delivery-agent workflow
- Barcode or QR-based shipment scanning
- Route and delivery-zone optimization
- Payment gateway integration
- Advanced service-level analytics
- Predictive delay monitoring
- Multi-language customer communication
- Integration with external logistics and accounting systems

11. APPENDIX

Appendix A – Core CMS Objects

Object	**Primary Purpose**
Customer__c	Customer master information
Shipment__c	Shipment booking and lifecycle
Delivery_Agent__c	Delivery workforce information
Branch__c	Branch and location information
Delivery_Update__c	Tracking and delivery events
Invoice__c	Shipment billing information

Appendix B – Key Automation

Automation	Purpose
Shipment validation rules	Prevent invalid operational values
Delivery status Flow	Synchronize shipment status from delivery updates
Reports	Provide filtered operational information
Dashboard	Present key courier metrics in one view

Appendix C – Suggested Test Data

Data Type	Sample
Customer	Retail customer with phone and email
Shipment	Domestic parcel with standard service
Delivery Agent	Active agent assigned to a branch
Branch	Main city branch
Delivery Update	In Transit / Out for Delivery / Delivered
Invoice	Shipment-linked invoice with payment status

Thank You